Climate & Us

50 years ago, Columbia University Professor and National Medal of Science Laureate Wally Broecker (1931-2019)

wrote a paper in *Science* magazine entitled "Climatic Change: Are we on the brink of pronounced global warming?" (W. Broecker 1975)

Compared to twentieth century average to that date:

1965-1975: exactly equal to the long-term average

1975 mean global temperature: +0.02^{o}C

Yet Broecker stated in the article:

"the exponential rise in the atmospheric carbon dioxide content will tend to become a significant factor and by early in the next century will have driven the mean planetary temperature beyond the limits experienced during the last 1000 years."

Global Warming

↓

Climate Change?

Frank Luntz, in 2001, in a memo to George W. Bush

"The scientific debate is closing [against us] but is not yet closed. There is still a window of opportunity to challenge the science."

"Voters believe that there is no consensus about global warming within the scientific community. Should the public come to believe that the scientific issues are settled, their views about global warming will change accordingly. Therefore, you need to continue to make the lack of scientific certainty a primary issue in the debate."

"The phrase "global warming" should be abandoned in favor of "climate change".

"A … story, even if factually inaccurate, can be more emotionally compelling than a dry recitation of the truth."

Global Warming

What We Know and What We Don't Know

Facts, Physics, Feedback, Fictions, Foreshadowing

Global Warming
of the
CLIMATE

Weather: *changes in minutes to years*
storms, fronts, droughts, cold snaps
Do I take an umbrella today?
Do I buy a warmer parka this winter?

Climate: *changes in decades to millennia*
ice ages, Medieval warm period
Do I buy a beachfront home?
Do I buy farmland in Canada?

What determines a planet's temperature?

What determines a planet's temperature?

Physics

Energy IN = Energy OUT

Chapter 1: Astronomical factors

What determines a planet's temperature?

Physics

Energy IN = Energy OUT

- ★ Sun's Energy Output
- Distance to Earth
- Tilt of Earth's Axis
- Reflectivity of Earth
  - Cloud cover
  - Ice cover
  - Vegetation
  - Aerosols

Composition of atmosphere

- Water Vapor
- Carbon Dioxide
- Methane
- Nitrous Oxides
- Chlorofluorocarbons
- Aerosols

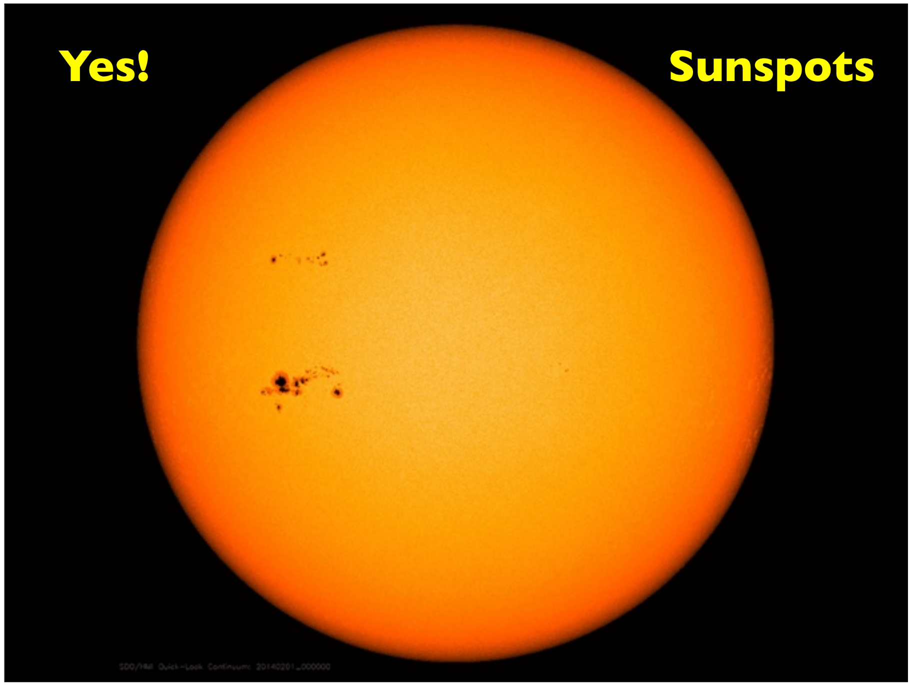
Sun's Energy Output

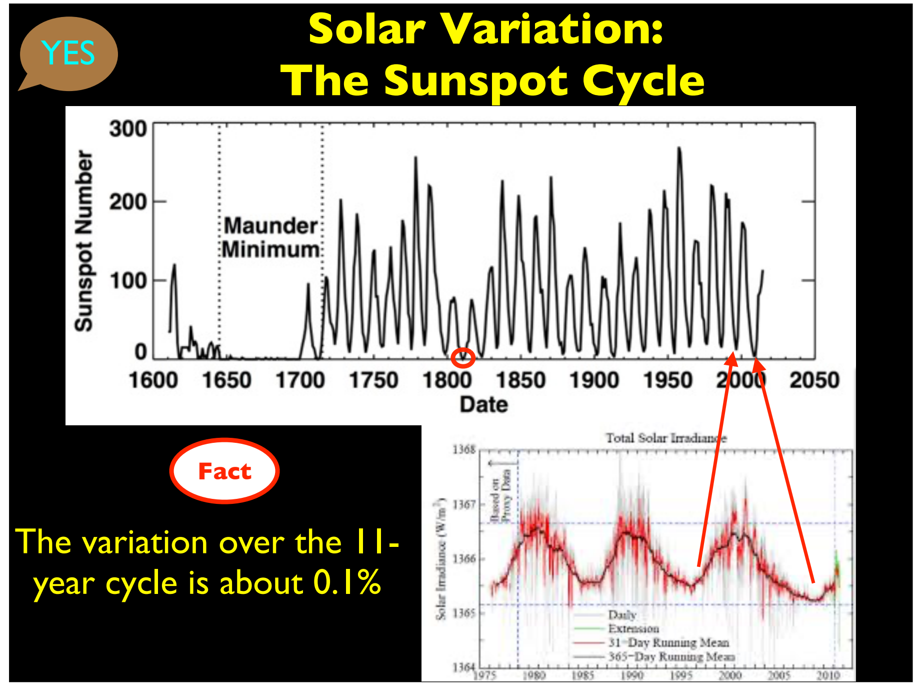
Fact

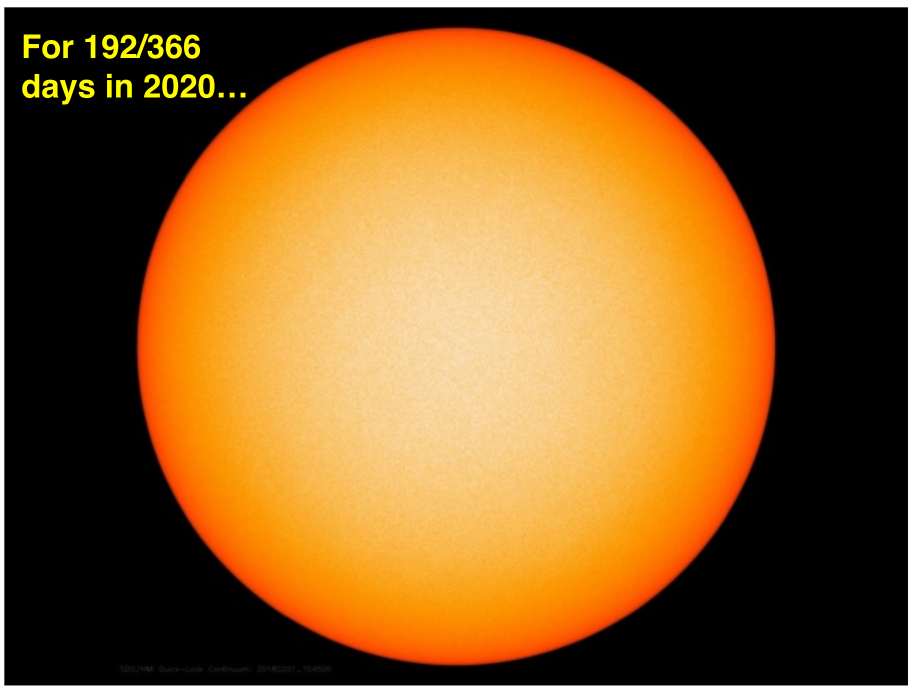
3.839×10^{26} Watts

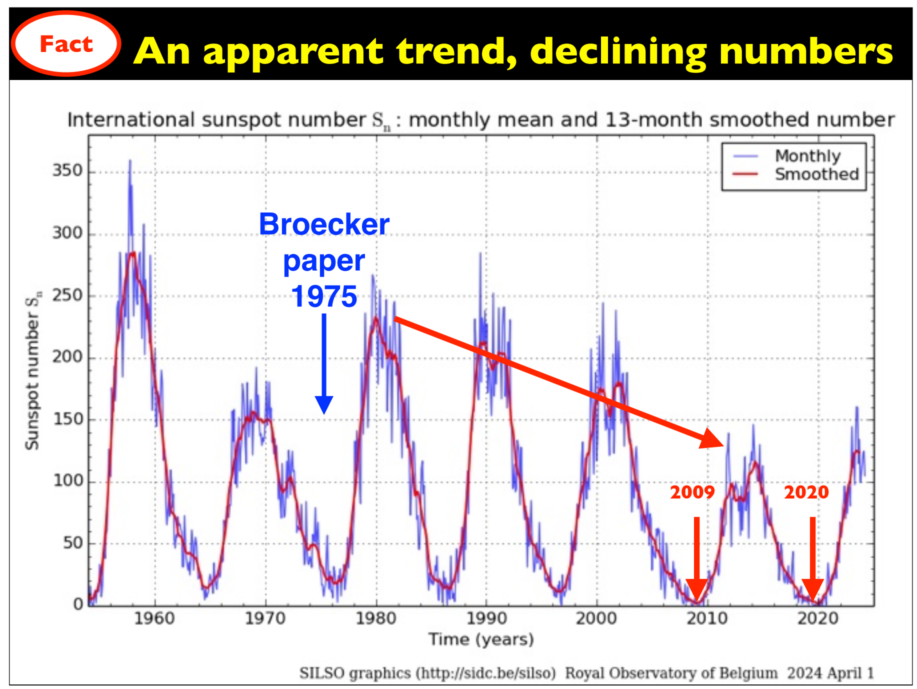
(383,900,000,000,000,000,000,000,000 Watts)

Does it change?

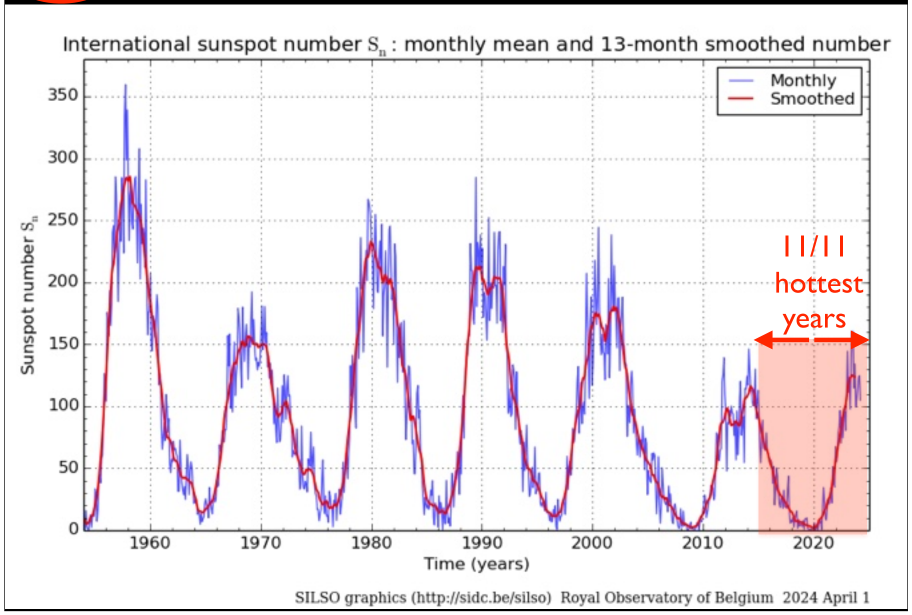

Yes!
Sunspots


YES
Solar Variation:
The Sunspot Cycle
Sunspot Number
300
200
100
0
Maunder
Minimum
1600
1650
1700
1750
1800
1850
1900
1950
2000
2050
Date
Fact
The variation over the 11-year cycle is about 0.1%
Total Solar Irradiance
Daily
Extension
31-Day Running Mean
365-Day Running Mean

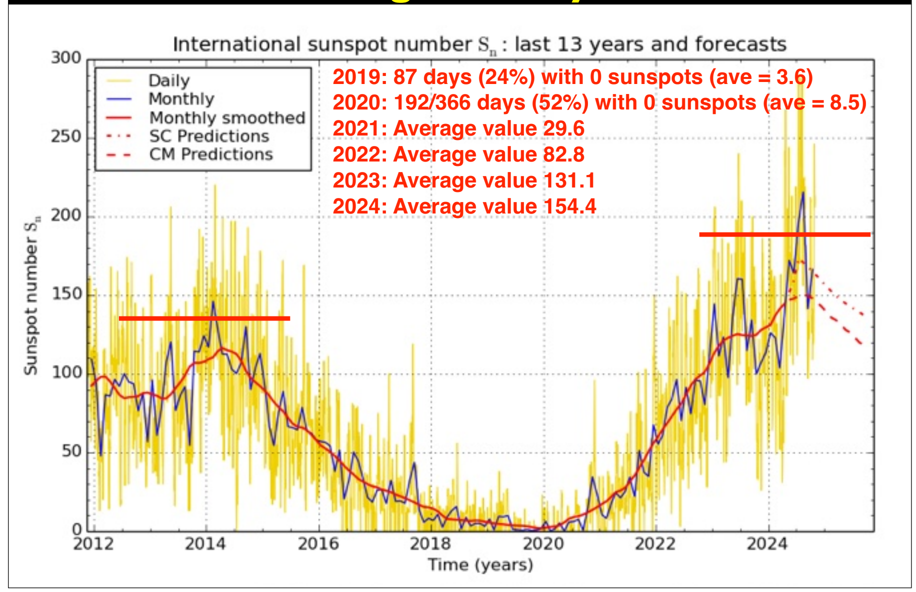

For 192/366 days in 2020…


Fact
An apparent trend, declining numbers
International sunspot number S_n : monthly mean and 13-month smoothed number
Monthly
Smoothed
Sunspot number S_n
350
300
250
200
150
100
50
0
Broecker paper 1975
2009
2020
1960
1970
1980
1990
2000
2010
2020
Time (years)
SILSO graphics (http://sidc.be/silso) Royal Observatory of Belgium 2024 April 1

Fact With spots at minimum, T was highest



Fact We're approaching the next maximum which is significantly more active



Fact

Average Total solar power: 174,400 TW (10^{12} W = 1 TW)

Other sources of energy?

- Moonlight 0.22 TW
- Planets and stars <0.1 TW
- Tides 3.3 TW
- Heat of formation 12-30 TW
- Radioactivity 15-41 TW

= 46+/-3

- Infalling matter <0.01 TW

TOTAL: ~50TW <0.03%

Changes: Sun +/- 0.1%; other <0.0006%

What determines a planet's temperature?

Physics

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√ Sun's Energy Output
★ Distance to Earth
Tilt of Earth's Axis
Reflectivity of Earth

- Cloud cover
- Ice cover
- Vegetation
- Aerosols

Composition of atmosphere

- Water Vapor
- Carbon Dioxide
- Methane
- Nitrous Oxides
- Chlorofluorocarbons
- Aerosols

Distance to Earth

Fact

$1.495978707 \times 10^{8}$ km

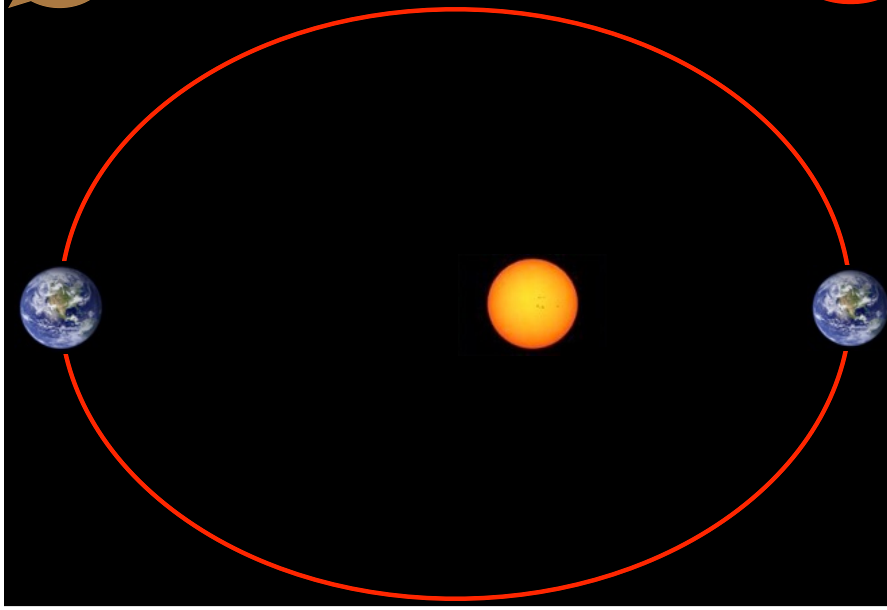
(~93,000,000 miles)

Does it change?

YES

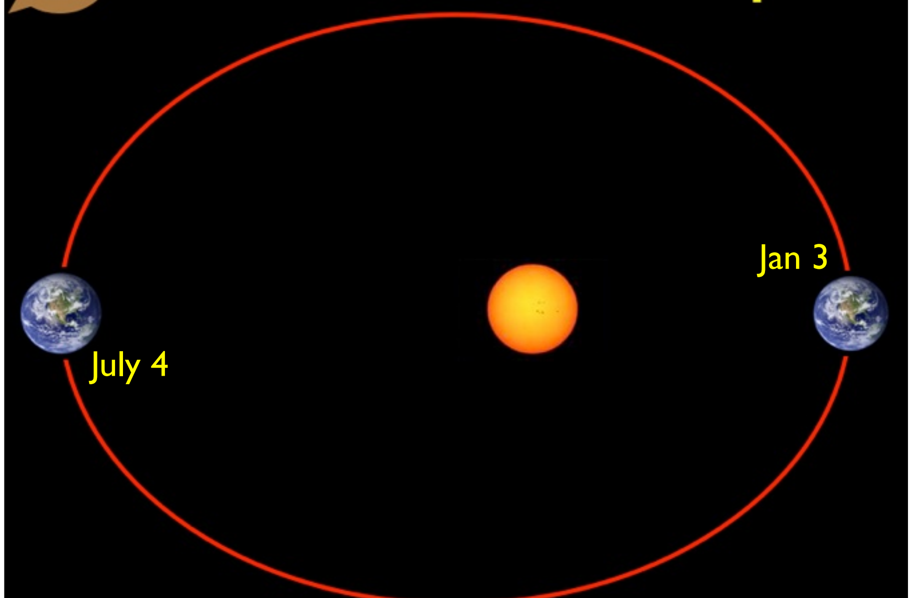
The Earth's orbit is an ellipse

Fact



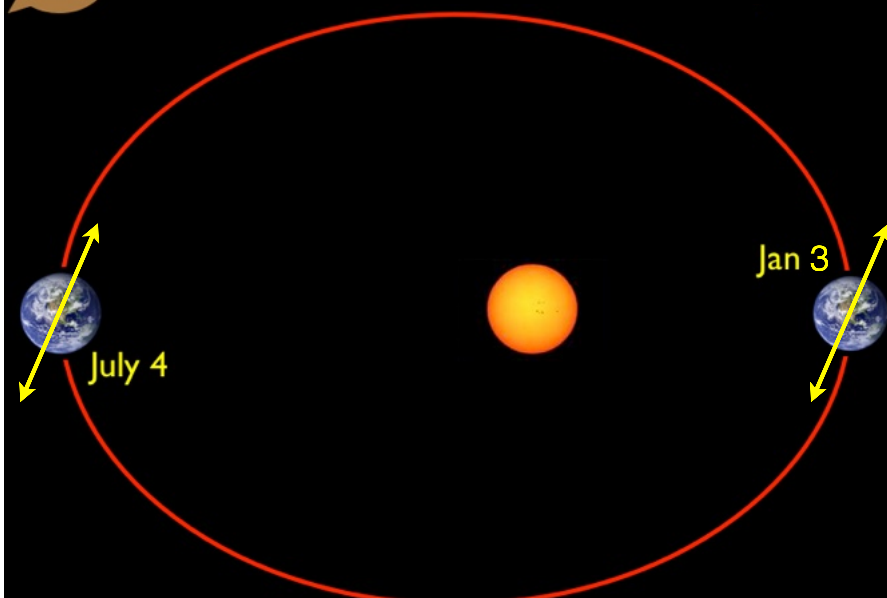
YES

The Earth's orbit is an ellipse



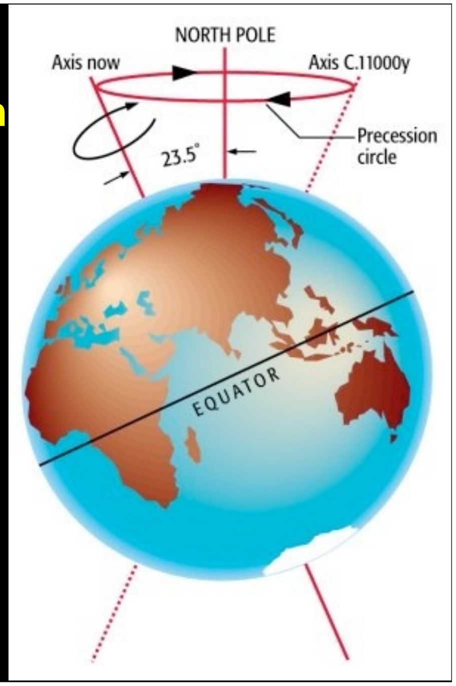
YES

The seasons result from axis tilt



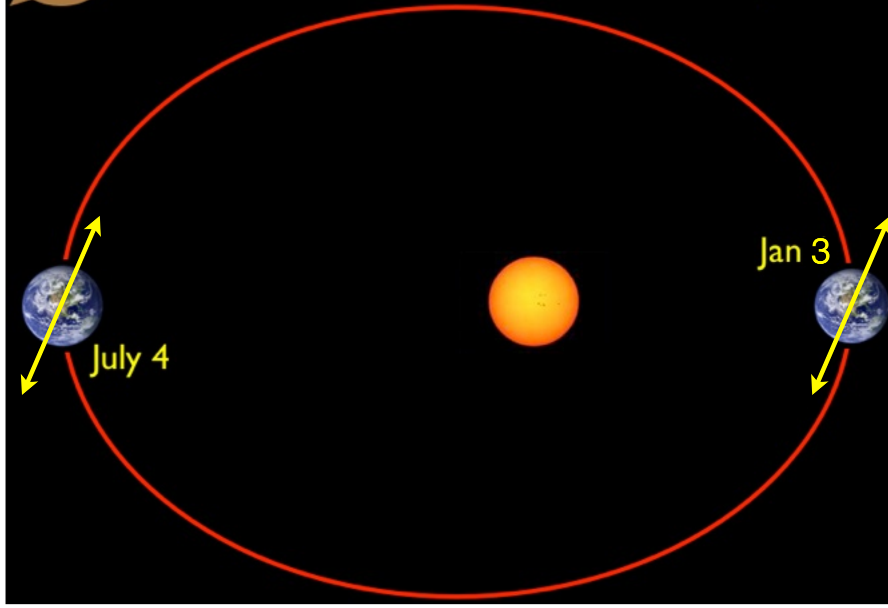
Physics

The Earth's rotation axis precesses

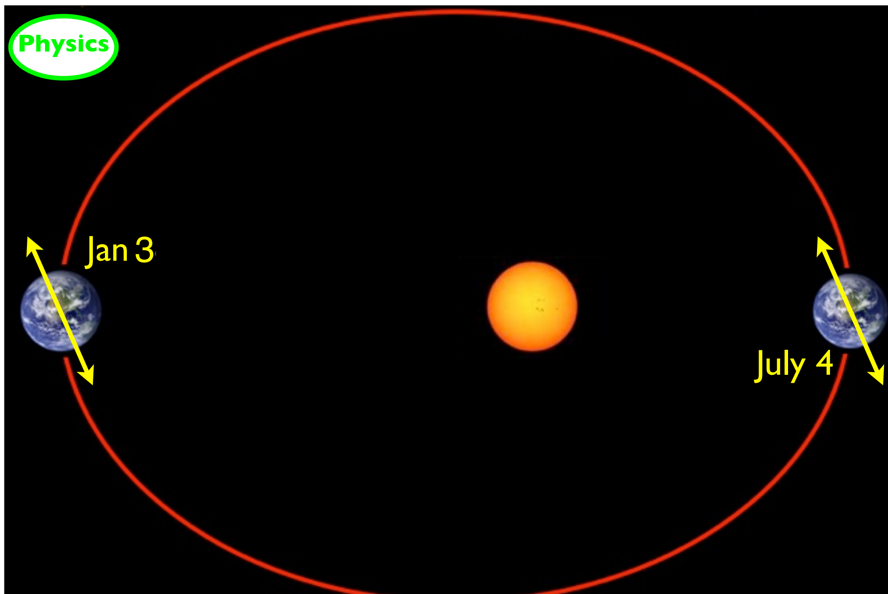


YES

The Seasons result from axis tilt

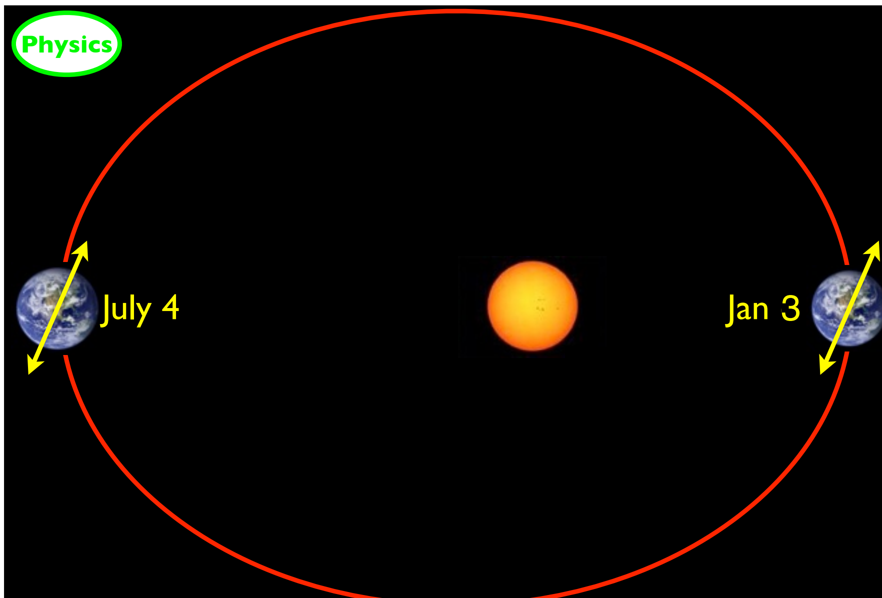


Physics



Hotter (but shorter) summers and colder (and longer) winters in the northern hemisphere

Physics



One cycle takes 20,000 - 23,000 years

What determines a planet's temperature?

Physics

Energy IN = Energy OUT

√ Sun's Energy Output
√ Distance to Earth
★ Tilt of Earth's Axis
Reflectivity of Earth
- Cloud cover
- Ice cover
- Vegetation
- Aerosols

Composition of atmosphere
- Water Vapor
- Carbon Dioxide
- Methane
- Nitrous Oxides
- Chlorofluorocarbons
- Aerosols

Tilt of Earth's Axis

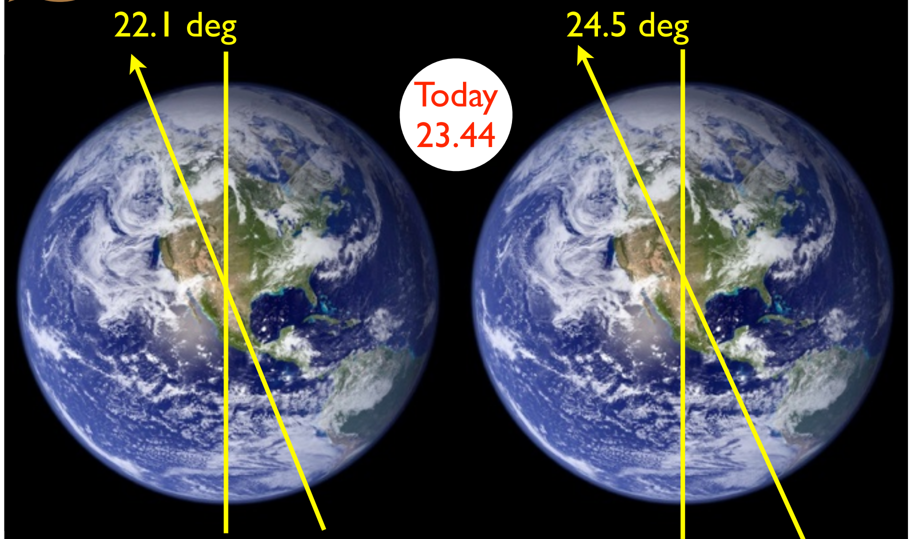
Fact

23.439281° pointed at Polaris

Does it change?

Tilt grows and shrinks

YES



One cycle takes 41,000 years

The ellipticity of the Earth's orbit

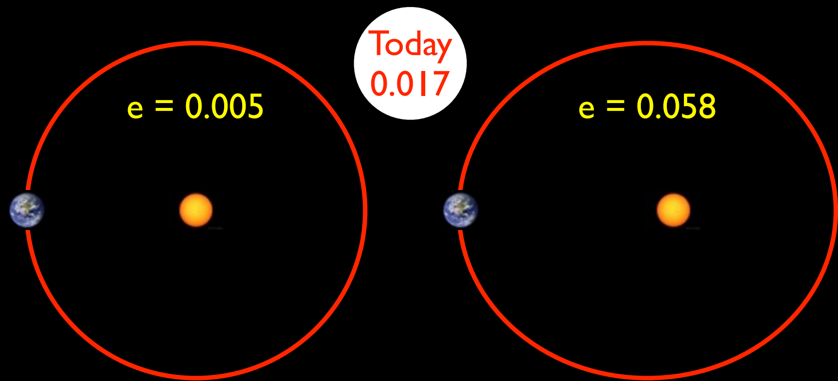
Fact

eccentricity = 0.0167

Does it change?

YES

Eccentricity grows and shrinks



One cycle takes 100,000 years

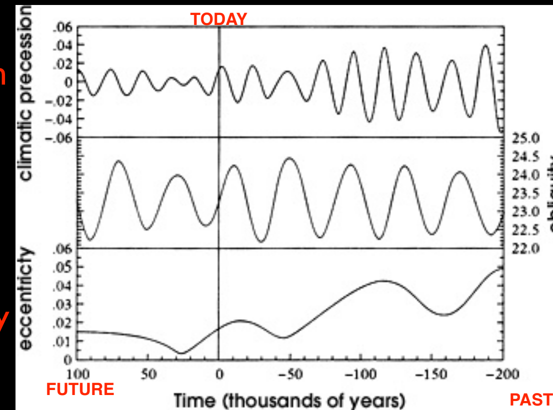
All three changes are going on at the same time

Physics

precession — 23,000 yrs

axis tilt — 41,000 yrs

eccentricity — 100,000 yrs



Milankovic Cycles

all three changes are COMPLETELY beyond our control

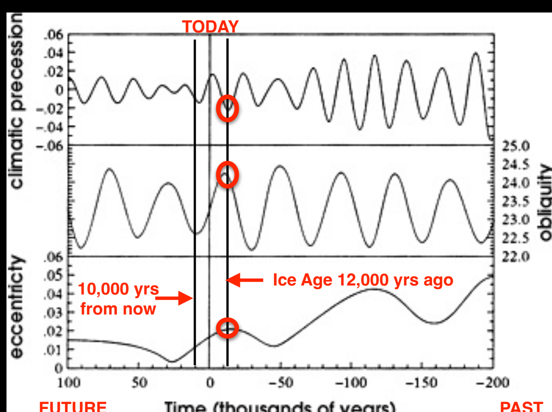
All three changes are going on at the same time

Physics

precession — 23,000 yrs

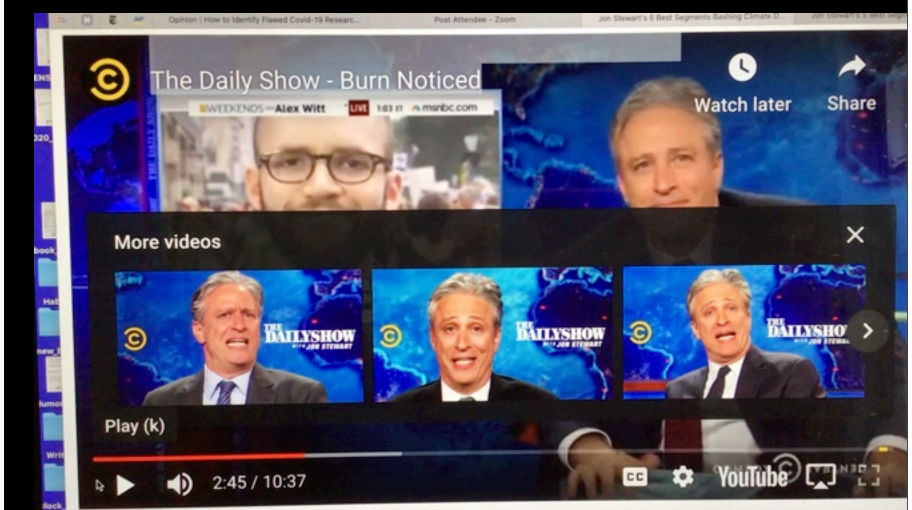
axis tilt — 41,000 yrs

eccentricity — 100,000 yrs



all three changes are COMPLETELY beyond our control

all three changes have a MAJOR impact on Earth's climate



But NONE of them is relevant to the rapid global warming we are experiencing on a timescale of decades

(maximum effect over 50 years is about 0.01° C)

Chapter 1: Astronomical factors

—The sun provides 99.97% of our energy
—It varies by 1 part in a thousand
—All other sources vary by < 6 parts per million
—Sun is quietest it has been in >100 yrs => cooler Earth

—The Earth-Sun distance changes over the year
—The Earth's orientation changes over ~23,000 yrs
—The Earth's axis tilt changes over ~41,000 yrs
—The Earth's orbital shape changes in ~100,000 yrs

All these changes affect Earth's climate
None of these changes is relevant to Global Warming
(in 50 years, their combined effect is ~0.01 degree)

Chapter 2: The Earth's surface

What determines a planet's temperature?

Physics

Energy IN = Energy OUT

√ Sun's Energy Output
√ Distance to Earth
√ Tilt of Earth's Axis
★ Reflectivity of Earth
- Cloud cover
- Ice cover
- Vegetation
- Aerosols

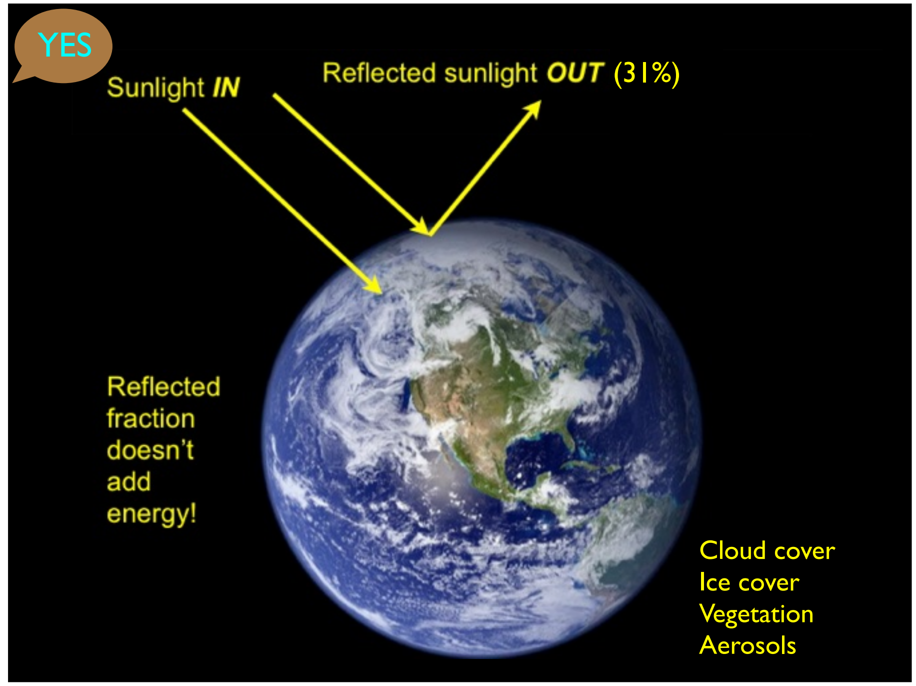
Composition of atmosphere
- Water Vapor
- Carbon Dioxide
- Methane
- Nitrous Oxides
- Chlorofluorocarbons
- Aerosols

Reflectivity of Earth

Fact

31% of light is reflected back to space

Does it change?



What determines a planet's temperature?

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★ Composition of atmosphere

- Water Vapor
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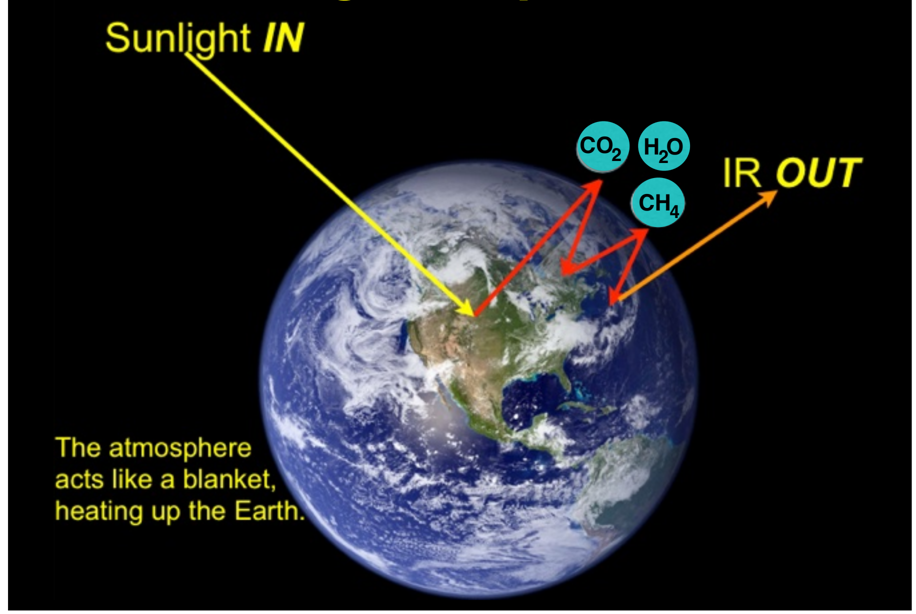
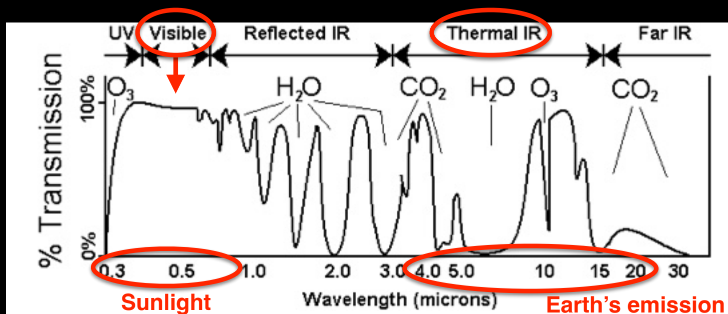
- Aerosols

A slight complication



Atmospheric transparency (and lack thereof) and the molecules responsible



λ_{max} = 0.0029 meters / T [Kelvins]

T_{Sun} = 5780 K => λ_{max} = 0.5 um; T_{Earth} = 289 K => λ_{max} = 10 um

Composition of the atmosphere; of 1 million atoms,

Facts

Nitrogen	778,885
Oxygen	208,736
Argon	9317
Water Vapor	~2500
Carbon Dioxide	427.6 (11/14/25)
Neon	18.2
Helium	5.24
Methane	1.93
Krypton	1.14
Hydrogen	0.50
Nitrous Oxides	0.337
Ozone	0.30
Chlorofluorocarbons	0.00012

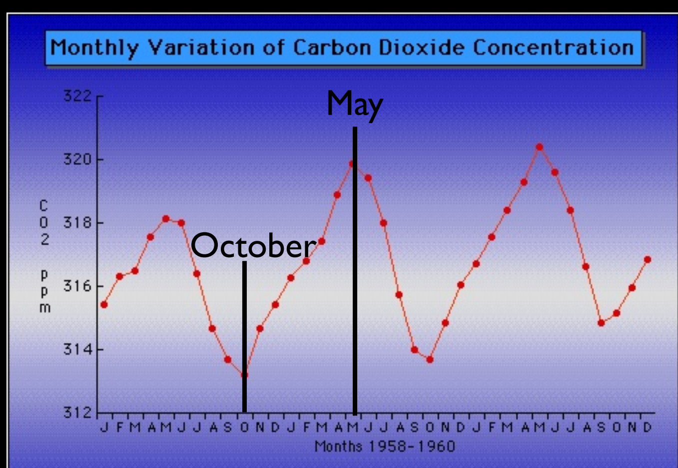
Greenhouse Gases in red

Does it change?

YES

Fact

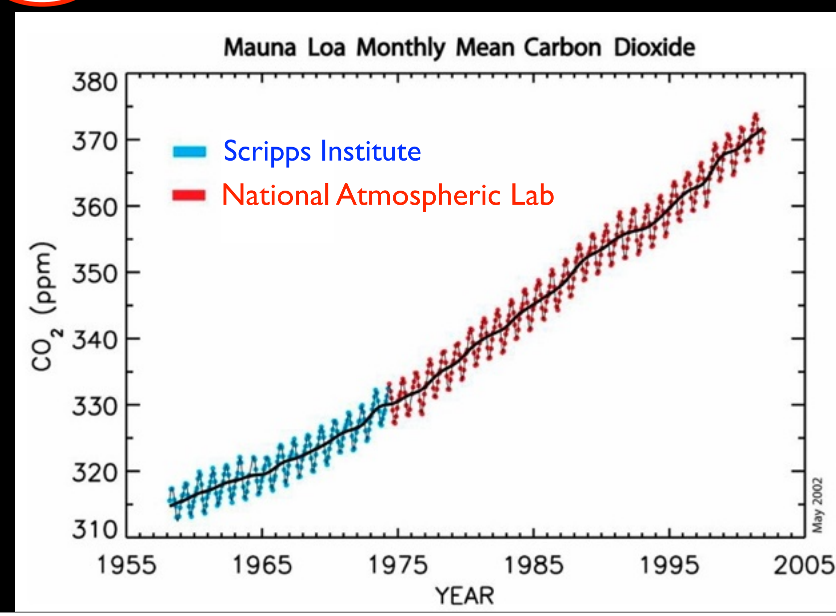
The Earth "Breathes" In and Out each Year

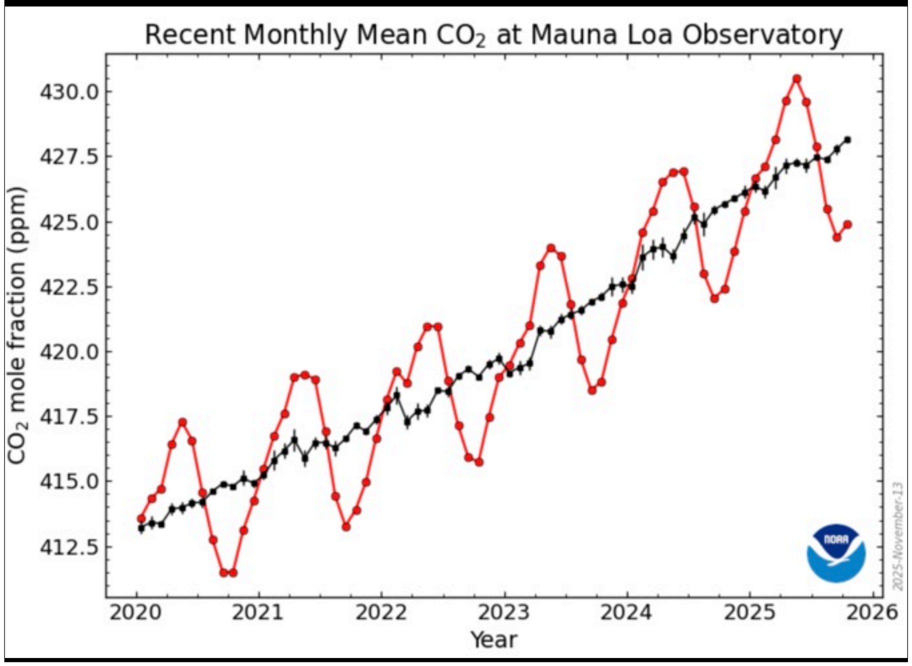


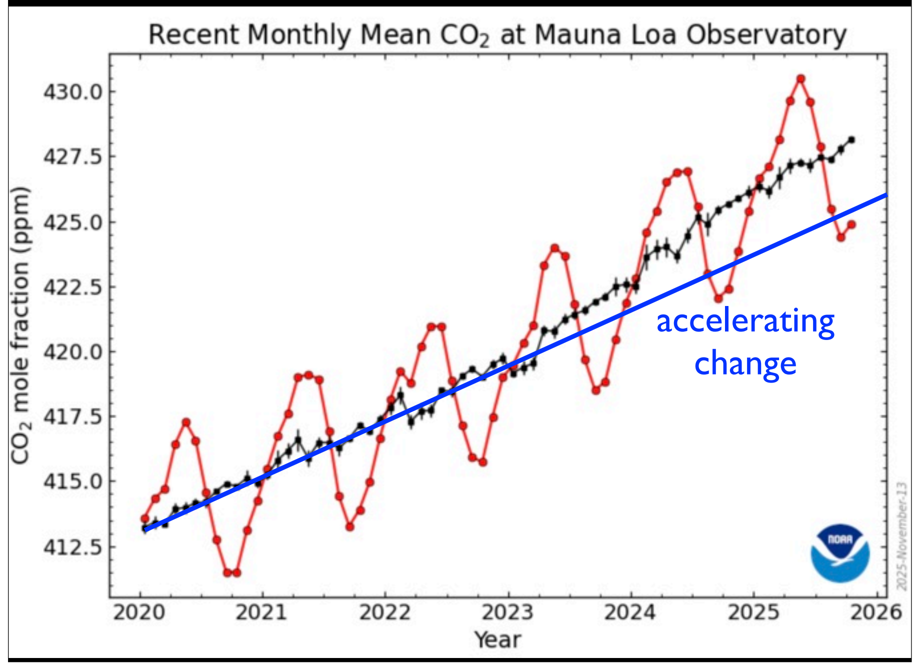
A 2% change between northern hemisphere spring and fall

Fact

But watch the long-term trend...

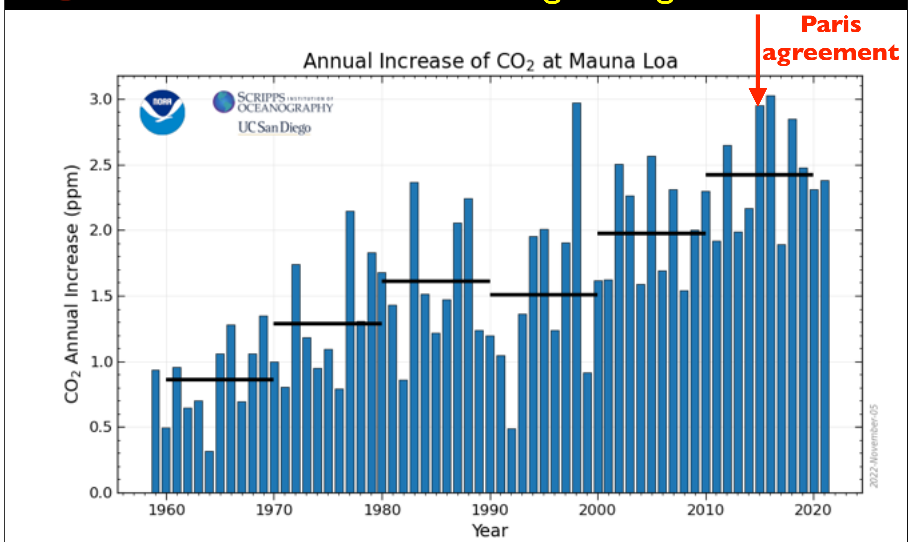






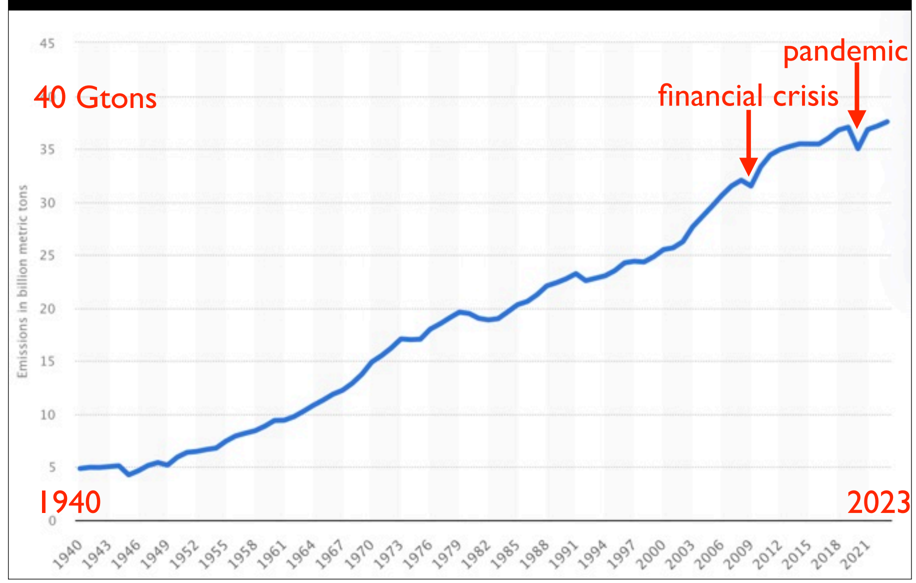
Fact

The increase is still growing…



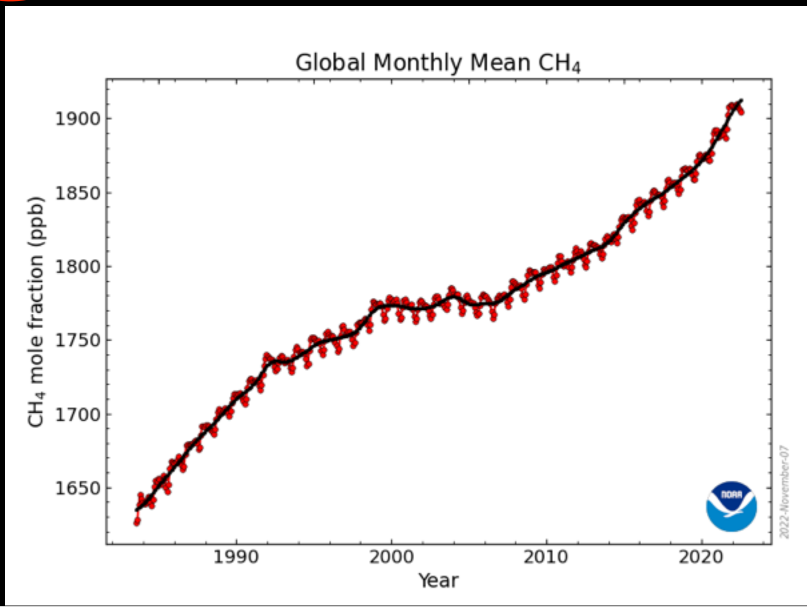
43% to air, 27% to plants, 30% to ocean

…because fossil fuel use is growing

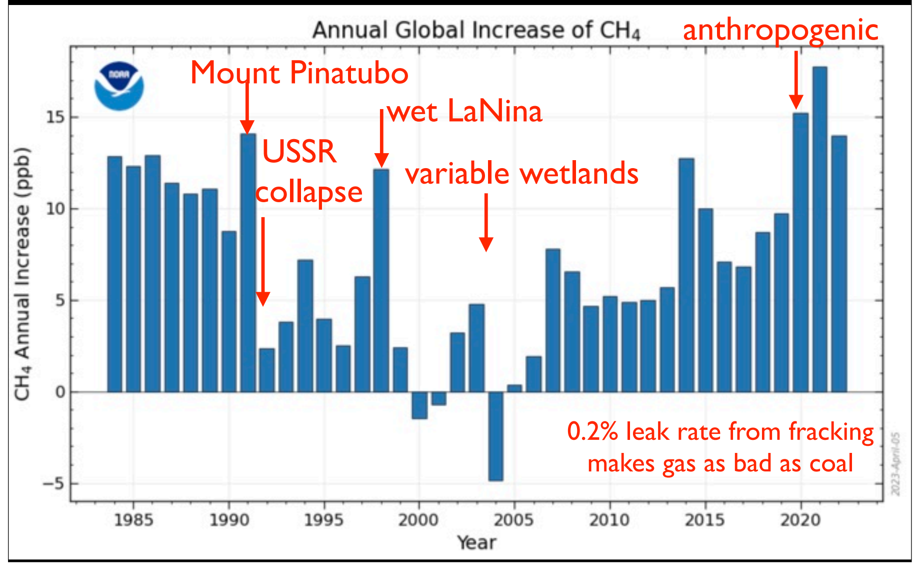


Fact

Methane abundance over time

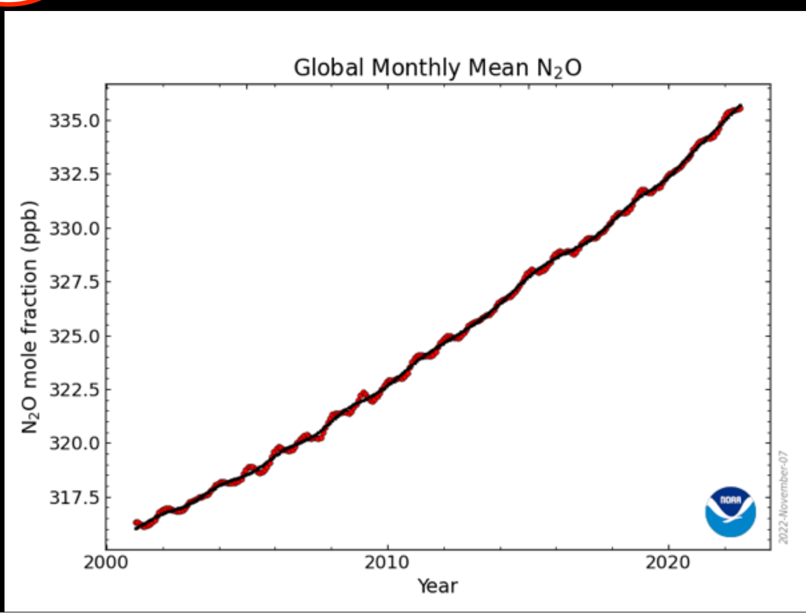


The increase is still accelerating

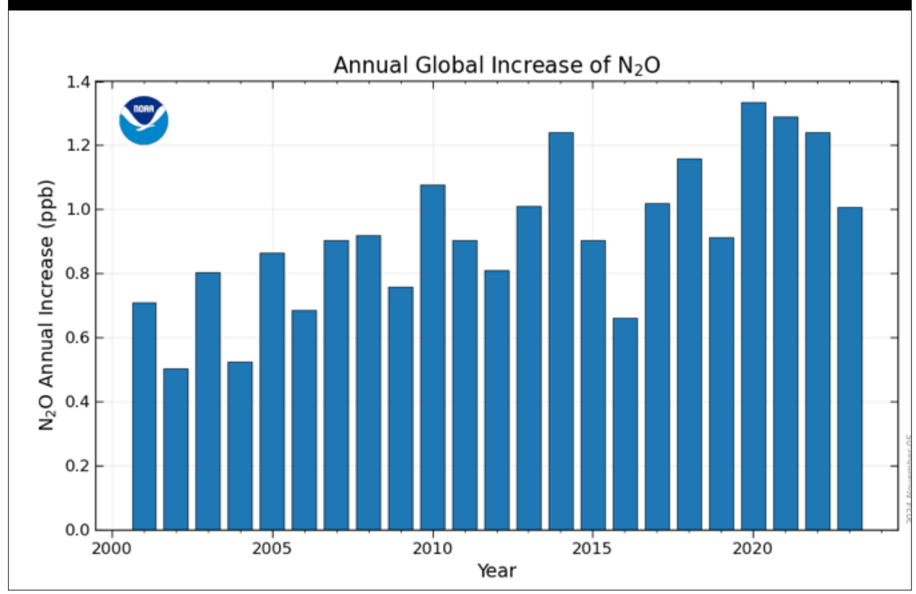


Fact

Nitrous Oxide abundance over time



The increase continues



Greenhouse Efficiencies:

Fact

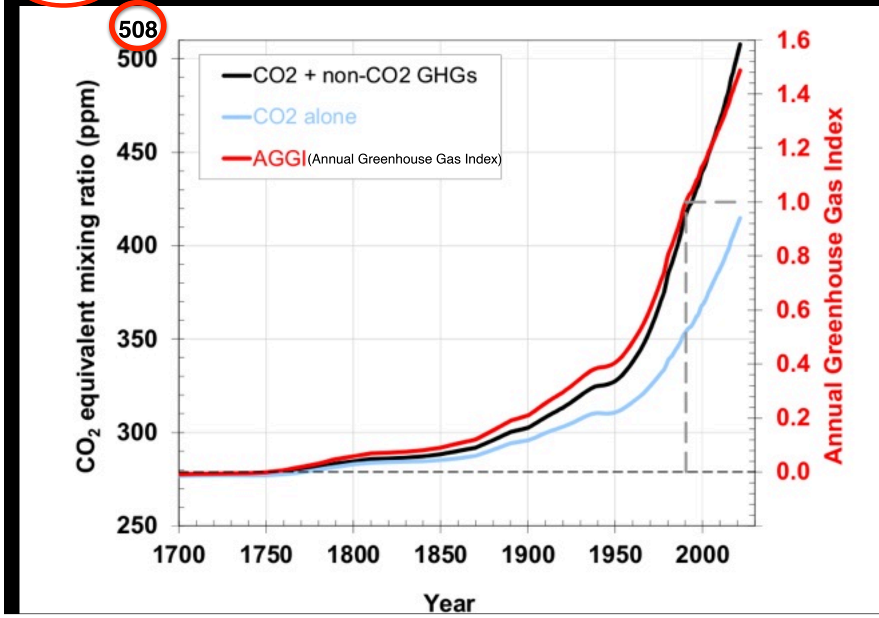
$H_2O = 0.15$, $CO_2 = 1.0$, $CH_4 = 26$, $N_2O = 215$

Gas	Pre-1750 tropospheric concentration[42]	Recent tropospheric concentration[43]	Absolute increase since 1750	Percentage increase since 1750	Increased radiative forcing (W/m²)[44]
Carbon dioxide (CO_2)	280 ppm[45]	420 ppm[46]	140 ppm	50%	2.2
Methane (CH_4)	700 ppb[47]	1915 ppb /[48]	1215 ppb /	275%	0.5
Nitrous oxide (N_2O)	270 ppb	336 ppb /[48]	66 ppb /	24%	0.2
Tropospheric ozone (O_3) +CFCs	237 ppb[42]	337 ppb[42]	100 ppb	42%	0.4

Water vapor is increasing about 1% / decade

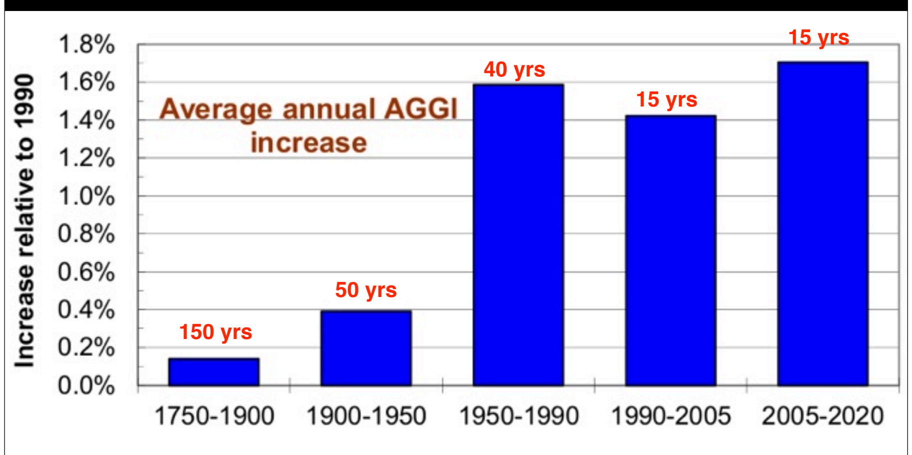
Accelerating growth in the burden

Fact



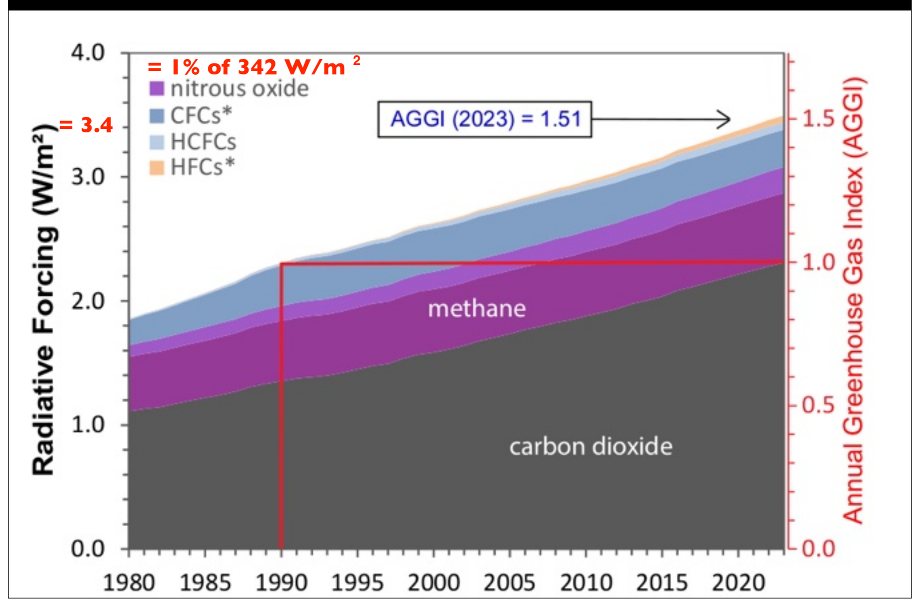
Fact

Accelerating growth in the burden

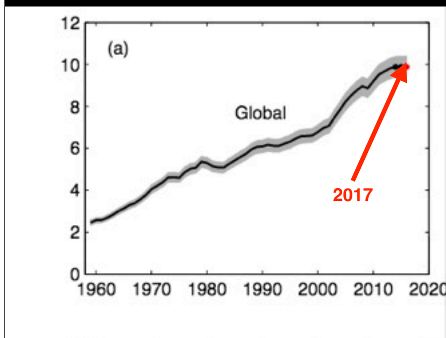


Fact

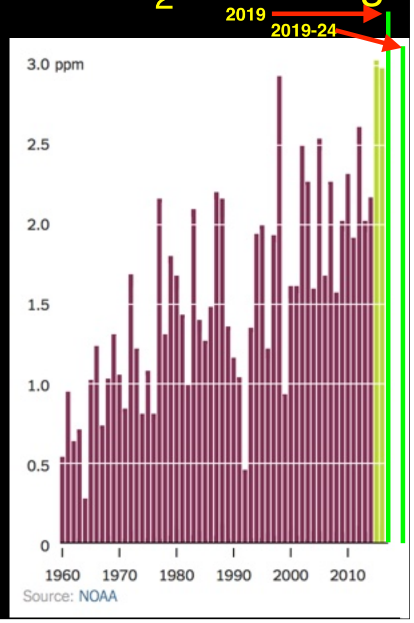
Total Greenhouse Gas burden



Emissions flattening

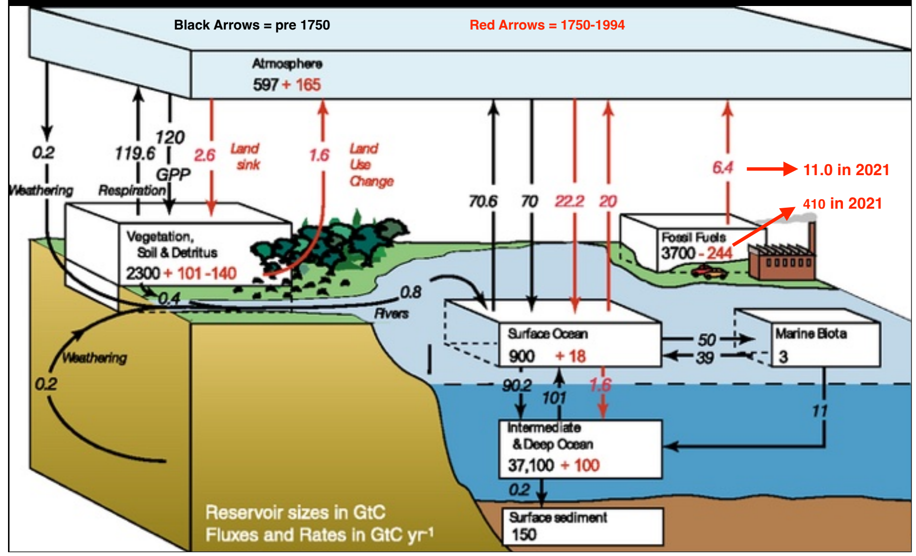


but CO_2 still rising

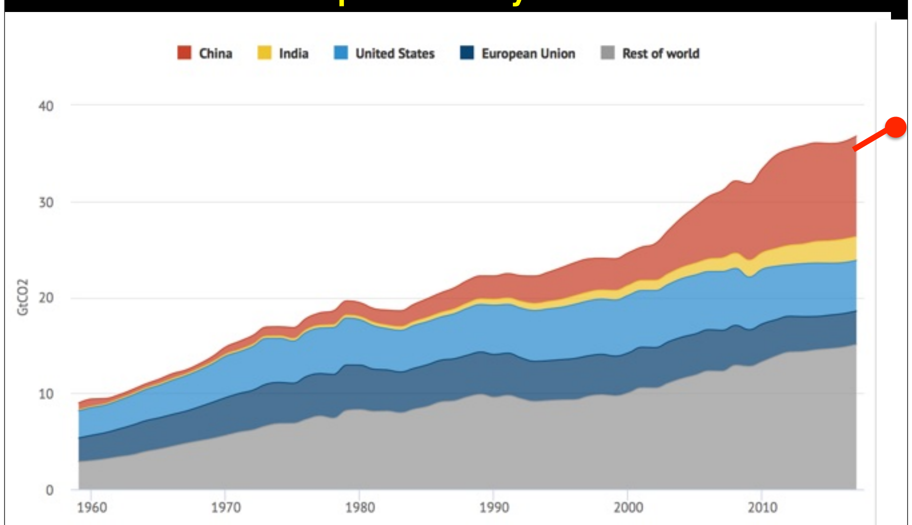


Are the sinks getting full to overflowing?
(ocean, trees, etc.)

Sources and Sinks over the past decade

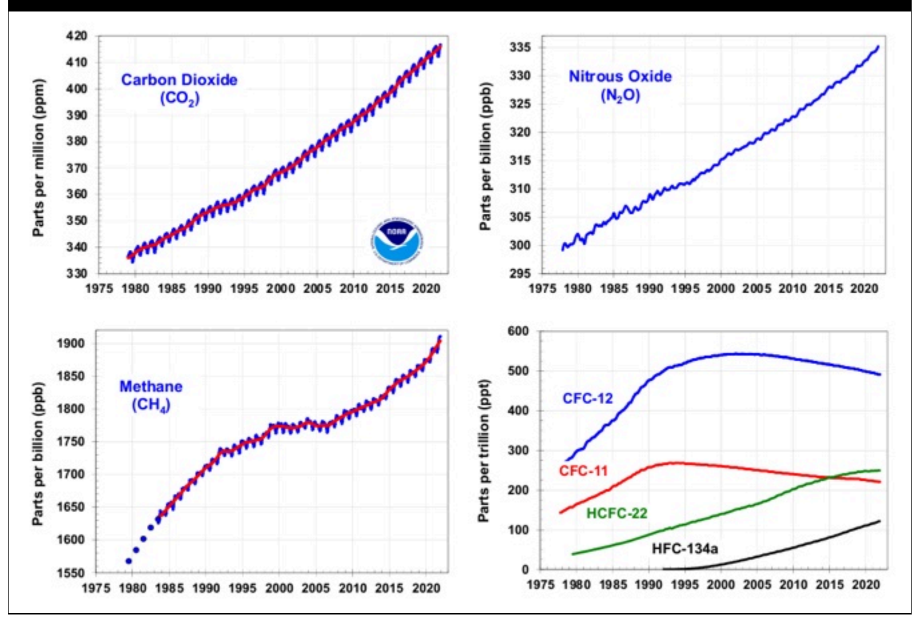


In 2017, the plateau ended with a 1.6% increase to 36.79 Gtons, a 0.6 Gt increase compared with 0.1Gt over the previous 3 years



2024: 37.4 Gt (+ 3% rise in 5 years) + 4.2Gt from land use changes

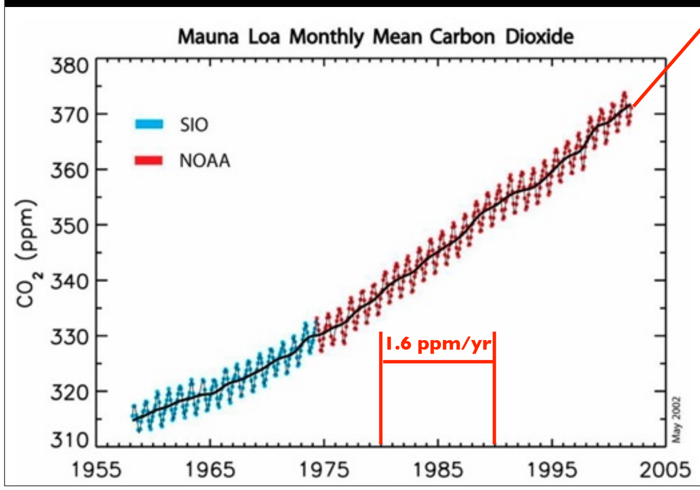
International agreements *can* work (CFCs)



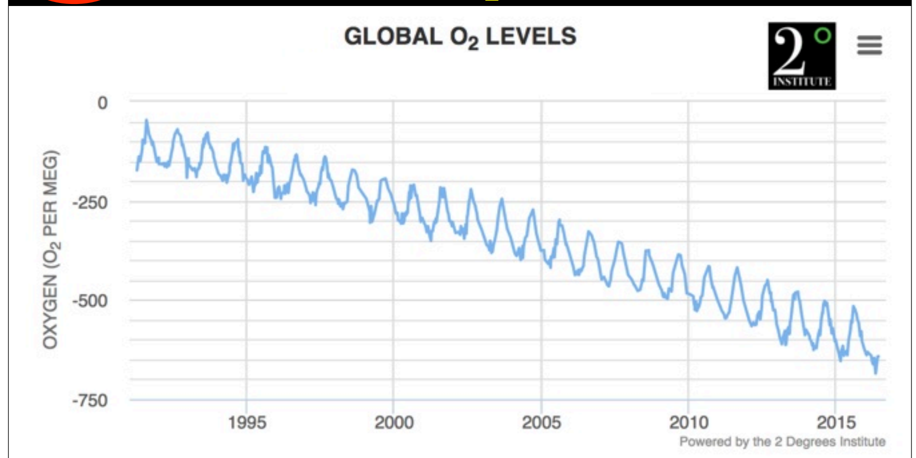
Fact

Nov 14, 2024 to
Nov 14, 2025
+4.1 ppm/yr

Everyone knows CO is increasing but…



Fact Measured rate of O_2 decline: 19ppm per yr



Fact Where is it going? Into CO_2 ! Maybe

Mass of missing oxygen ~ mass in new CO_2

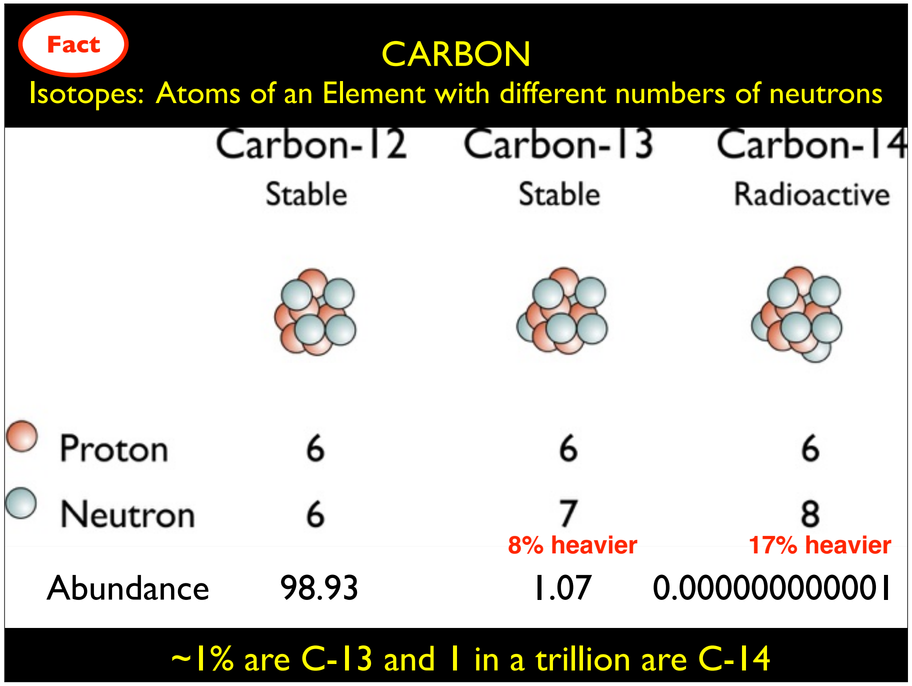
Chapter 2: The Earth's surface

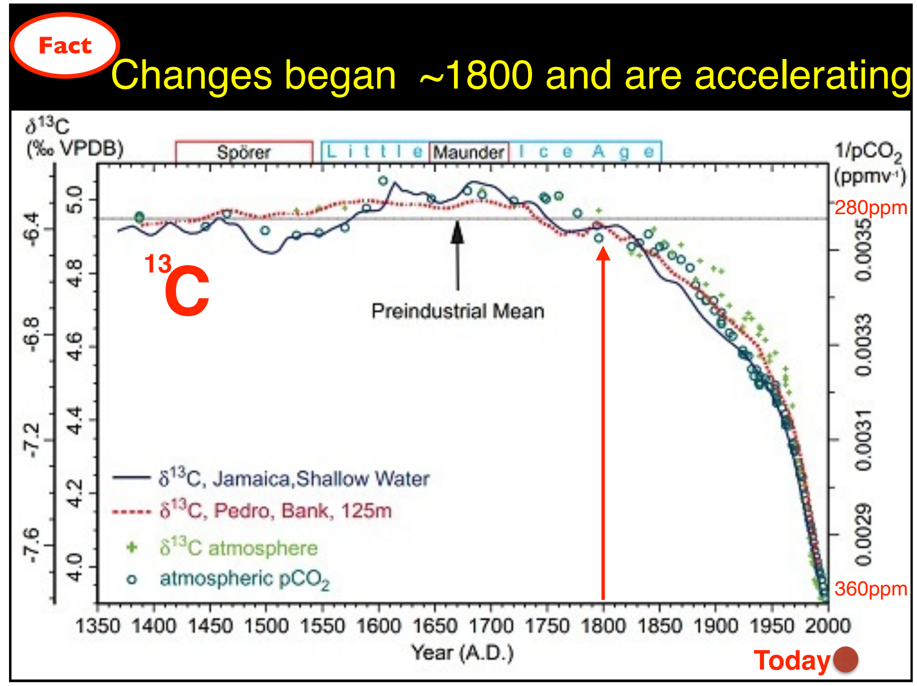
—~1/3 of sunlight is reflected
—the exact fraction is constantly changing

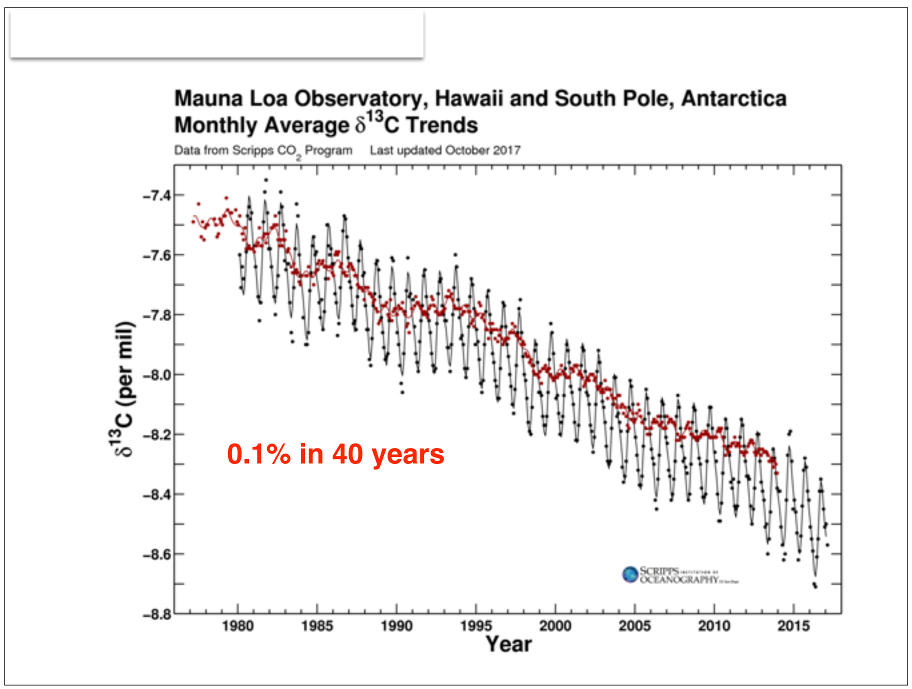
—incoming light is at ~0.5um; air is transparent
—outgoing light is at ~10um; air is partly opaque
—resulting Greenhouse effect keeps Earth comfortable
—too much Greenhouse effect warms the Earth

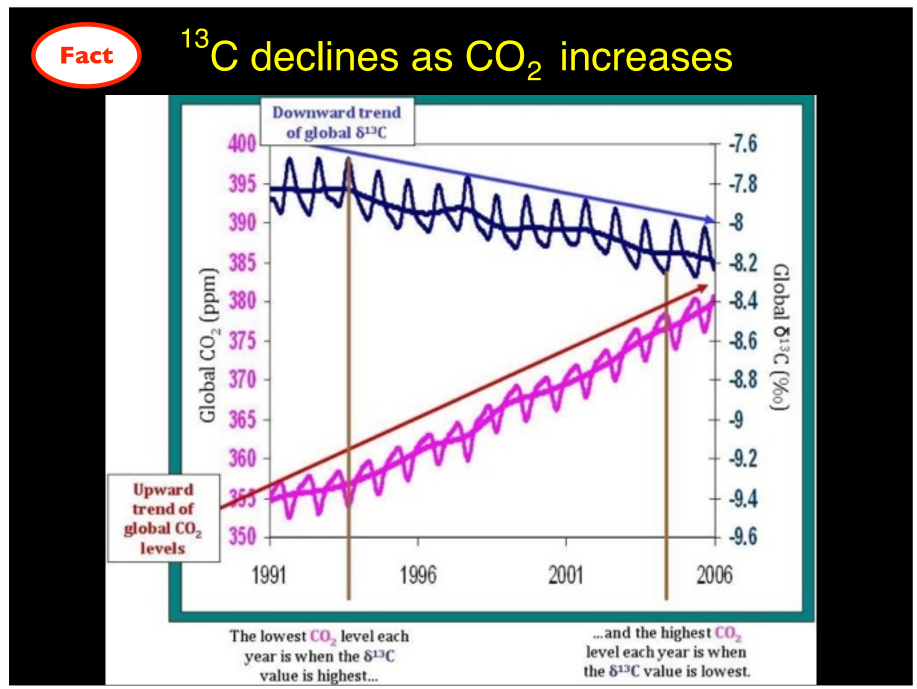
—Carbon dioxide (CO_2), Methane (CH_4) Water (H_2O)
Nitrous Oxide (N_2O), Ozone (O_3) all warm Earth
—all the Greenhouse gases are increasing
—Oxygen (O_2) is declining —> CO_2?

Chapter 3: The source of CO_2

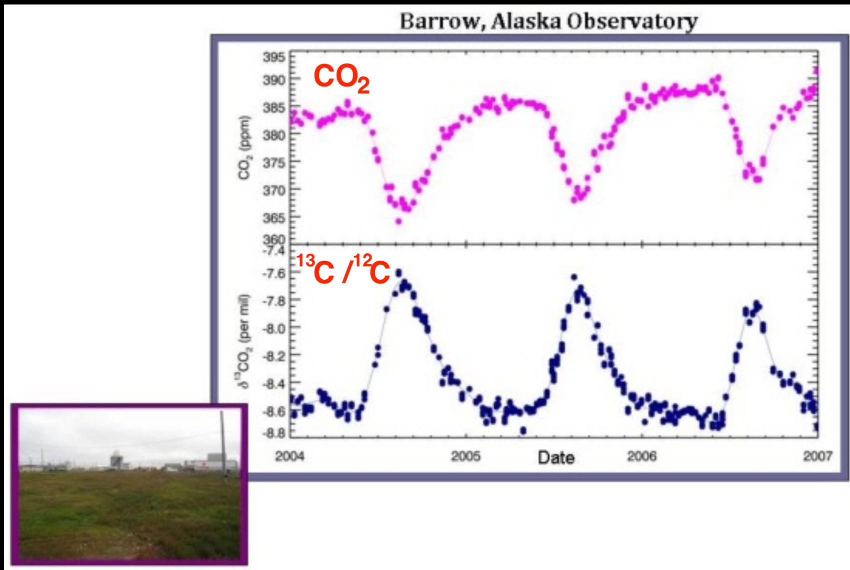

Fact
CARBON
Isotopes: Atoms of an Element with different numbers of neutrons
Carbon-12
Stable
Carbon-13
Stable
Carbon-14
Radioactive
Proton
6
6
6
Neutron
6
7
8
8% heavier
17% heavier
Abundance
98.93
1.07
0.0000000000001
~1% are C-13 and 1 in a trillion are C-14


Fact
Changes began ~1800 and are accelerating
δ13C (‰ VPDB)
Spörer
Little Maunder Ice Age
1/pCO2 (ppmv-1)
280ppm
360ppm
13C
Preindustrial Mean
δ13C, Jamaica,Shallow Water
δ13C, Pedro, Bank, 125m
δ13C atmosphere
atmospheric pCO2
Year (A.D.)
Today


Mauna Loa Observatory, Hawaii and South Pole, Antarctica
Monthly Average δ13C Trends
Data from Scripps CO2 Program Last updated October 2017
δ13C (per mil)
0.1% in 40 years
Year


Fact
13C declines as CO2 increases
Downward trend of global δ13C
Global CO2 (ppm)
Global δ13C (‰)
Upward trend of global CO2 levels
The lowest CO2 level each year is when the δ13C value is highest...
...and the highest CO2 level each year when the δ13C value is lowest.

^{12}C and ^{13}C are anti-correlated



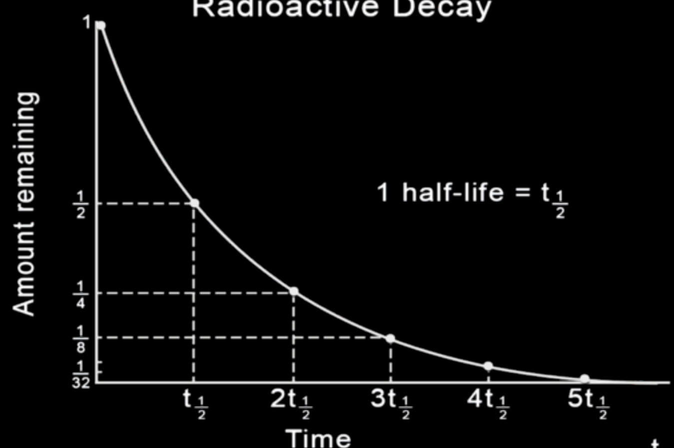
^{14}C is radioactive -- it transforms spontaneously to Nitrogen

C-14 Decay

$$^{14}_{6}C \longrightarrow {}^{14}_{7}N + {}^{0}_{-1}e + \overline{\nu}$$

Radioactive Decay

Physics

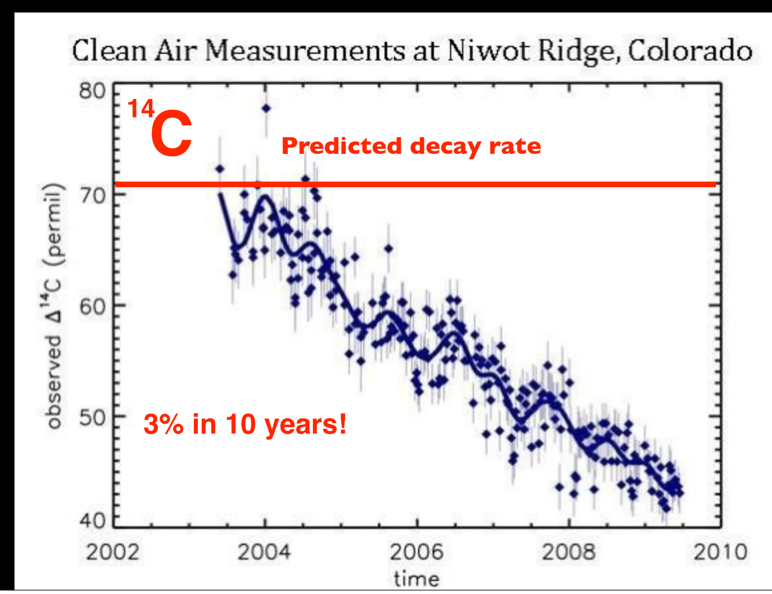


$$\text{Number remaining} = \text{Number at start} \times \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$$

For C-14, the half-life is 5730 years

The decrease in ^{14}C today

Fact



What does all this say about the source of the rapidly rising CO_2 content of Earth's atmosphere?

$\frac{^{13}C}{^{12}C}$ ratio

in CO_2

HIGH

MEDIUM

LOW

$\frac{^{14}C}{^{12}C}$ ratio

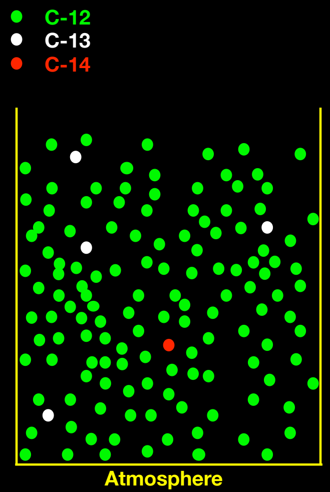
in CO_2

HIGH

MEDIUM

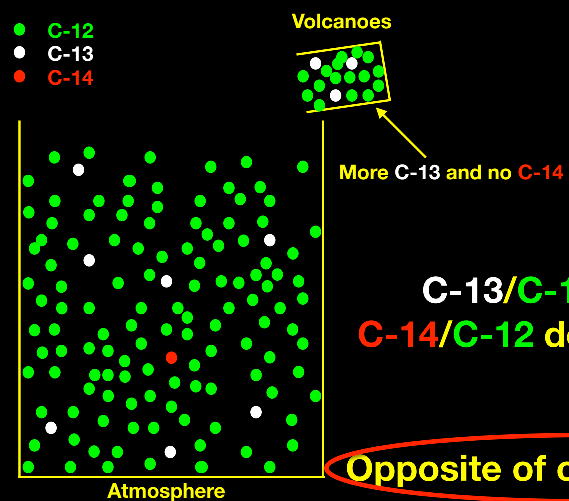
LOW (= 0)

So, here's the situation…



we are going to add some gas to the atmosphere that contains Carbon isotopes in different proportions

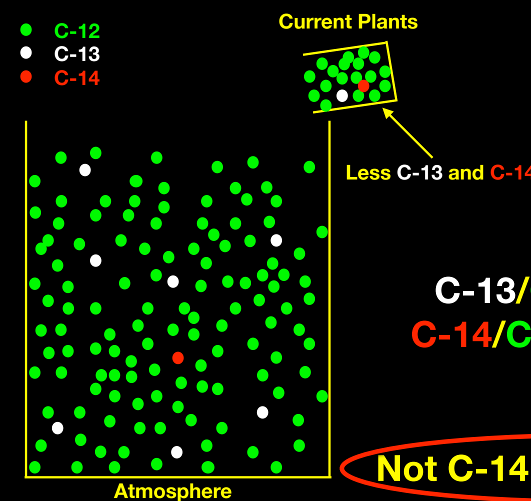
So, here's the situation…



C-13/C-12 rises and
C-14/C-12 declines slightly

Opposite of observed trend!

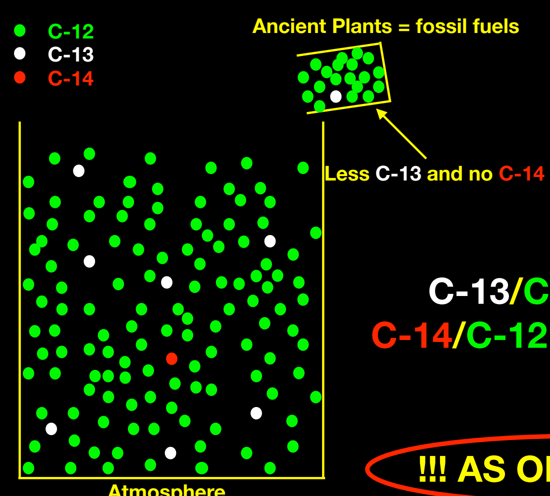
So, here's the situation…



C-13/C-12 falls and
C-14/C-12 stays small

Not C-14 observed trend!

So, here's the situation…



C-13/C-12 falls and
C-14/C-12 also decreases

!!! AS OBSERVED !!!

Amount **Sources** **New CO_2 in the air**

high

volcanoes

^{13}C atmosphere, ocean — declining C-13

recent plants, ancient plants ←

low

high

nuclear tests

^{14}C recent plants — declining C-14

ancient plants ← **= fossil fuels**

low

FACT

Earth's atmospheric CO_2 content is increasing at 0.8% per year and most of it (~90%) is a consequence of burning fossil fuels.

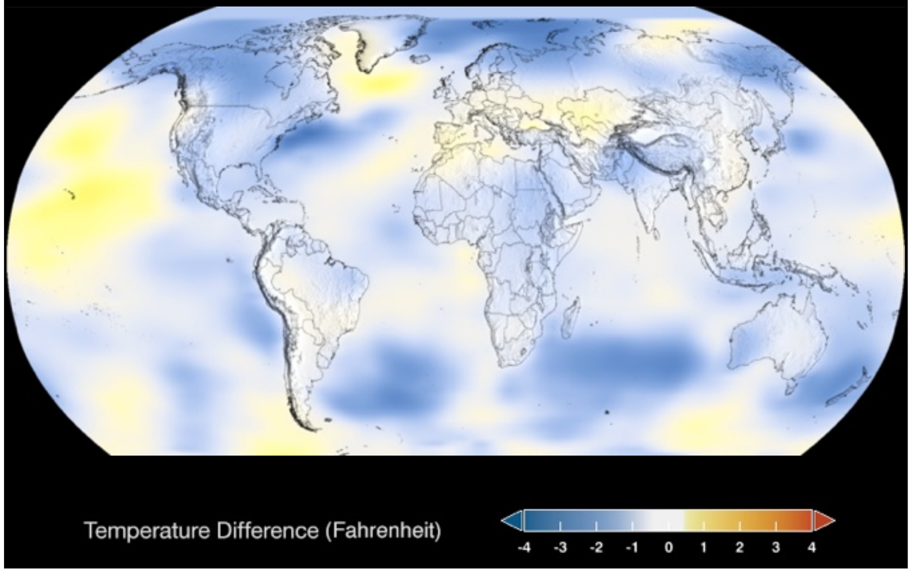
Chapter 3: The source of CO_2

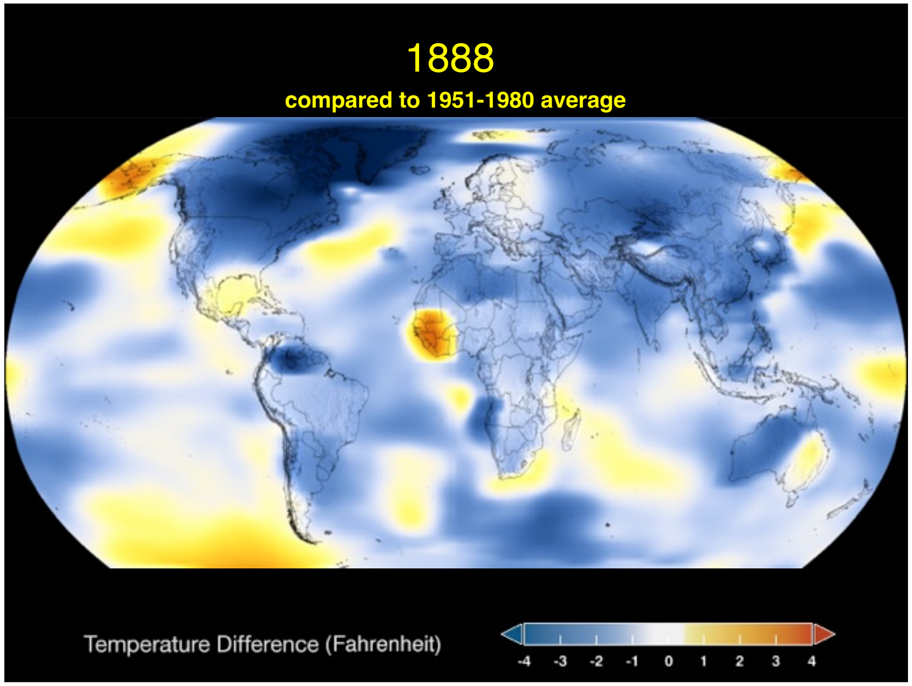
Carbon isotopes provide unambiguous evidence that the steady increase in CO_2 comes from burning fossil fuels

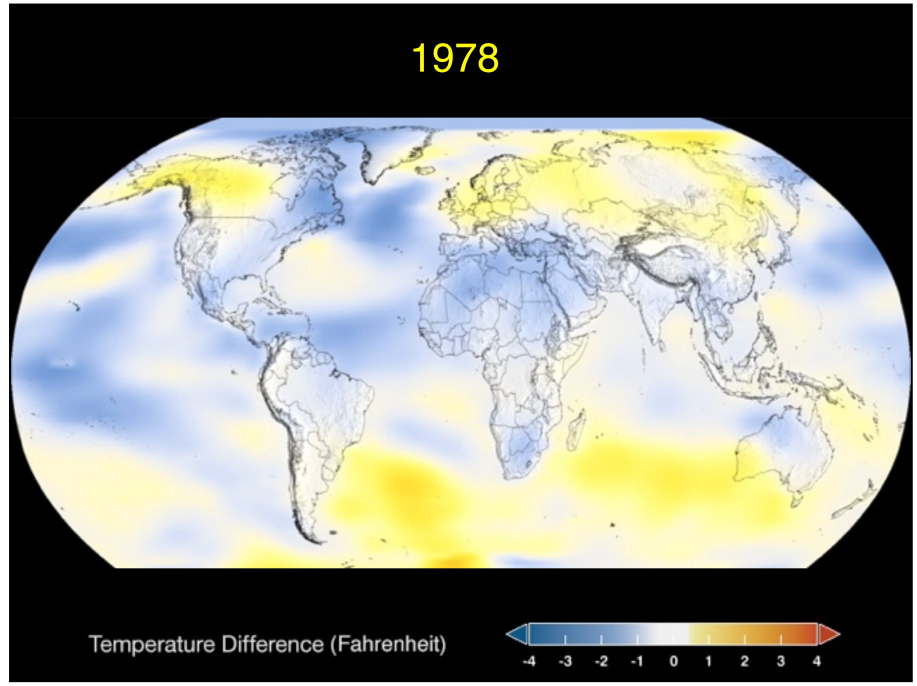
Chapter 4: The temperature record

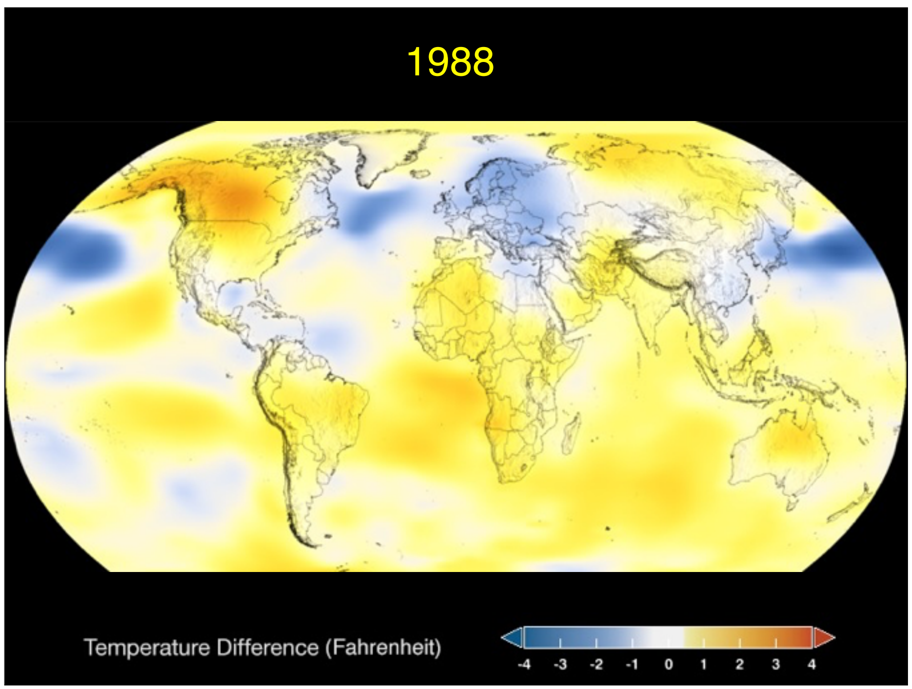
1968: Mean = 57.0F

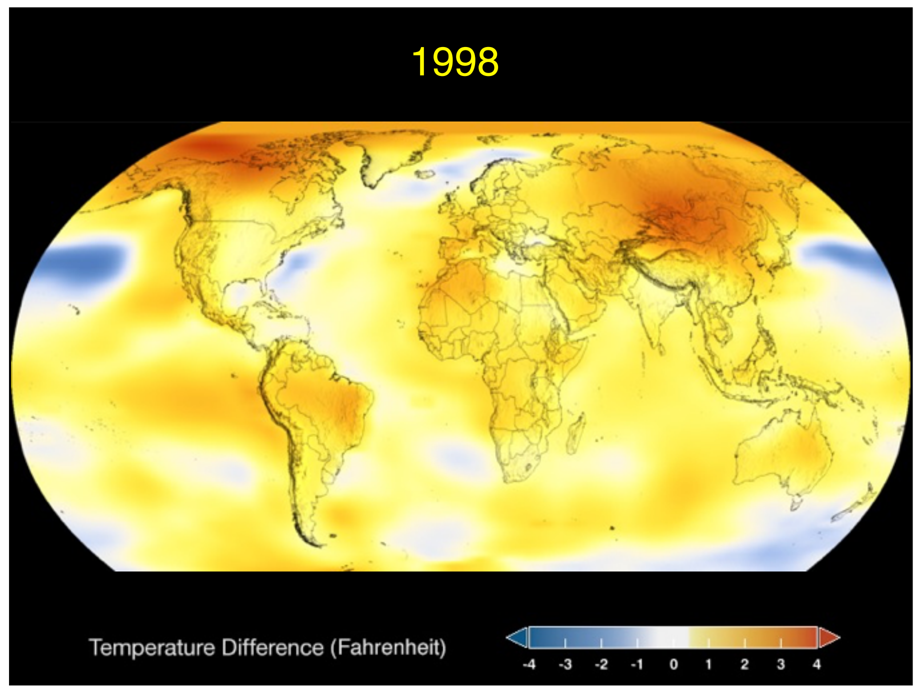
compared to 1951-1980 average

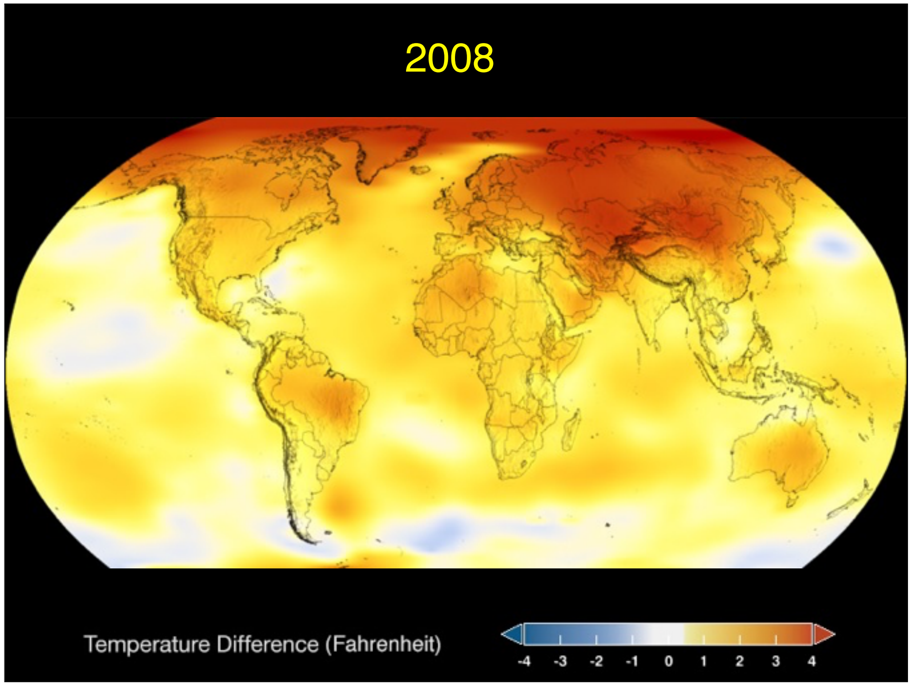


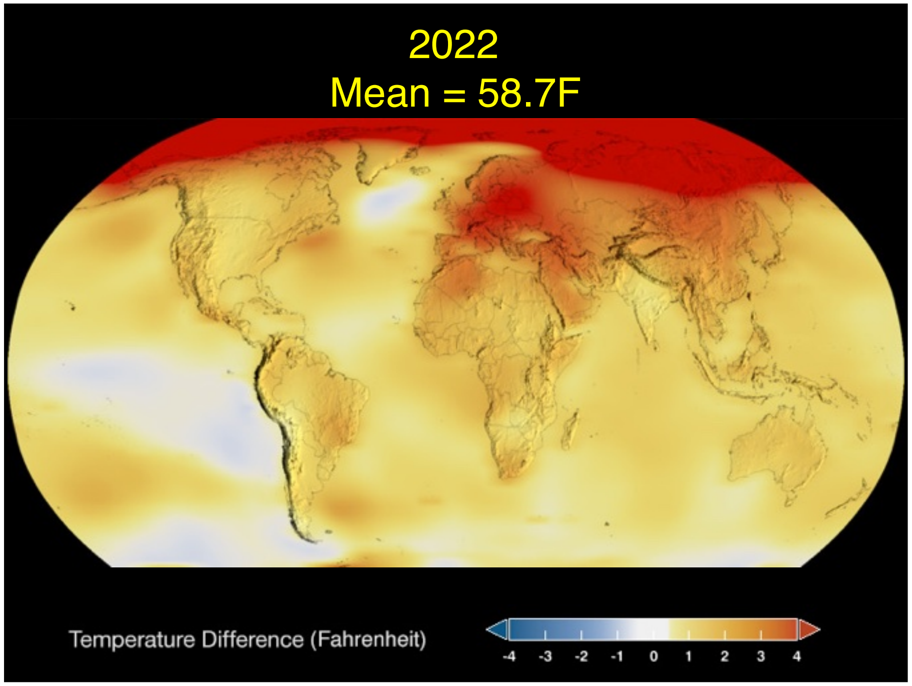

1888
compared to 1951-1980 average
Temperature Difference (Fahrenheit)
-4 -3 -2 -1 0 1 2 3 4

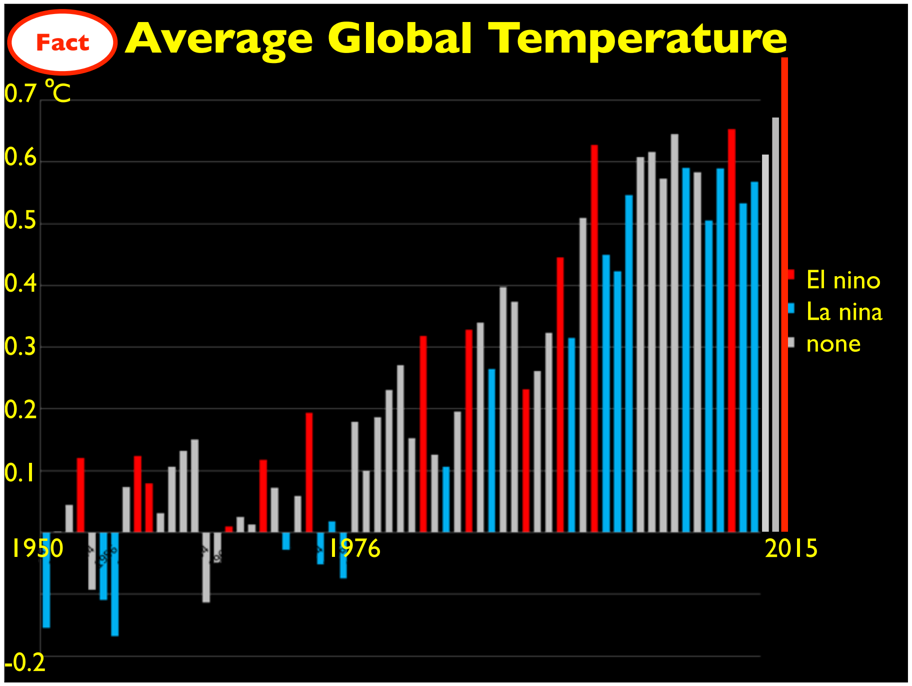

1978
Temperature Difference (Fahrenheit)
-4 -3 -2 -1 0 1 2 3 4

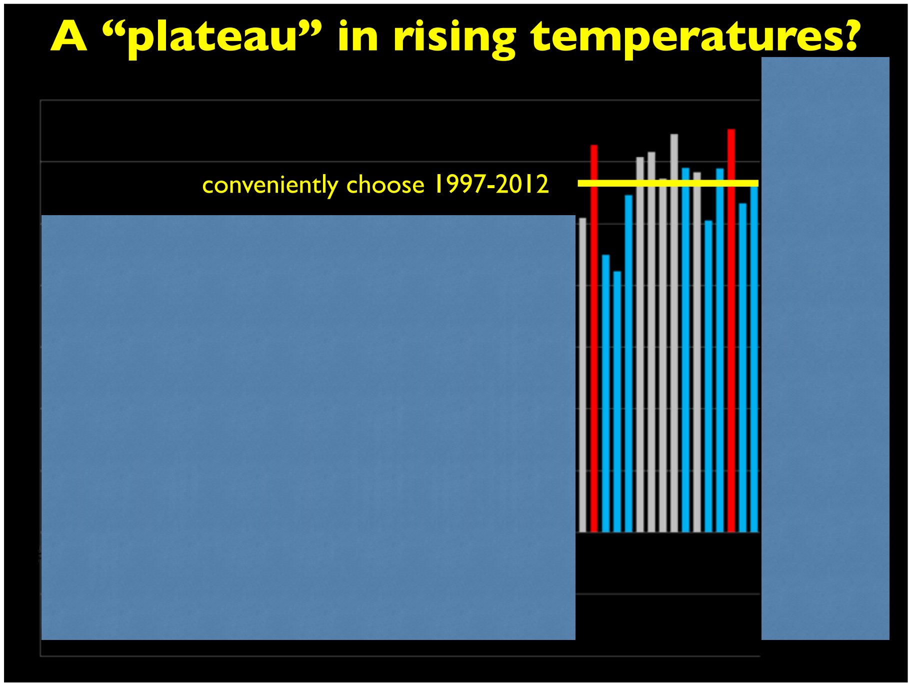

1988
Temperature Difference (Fahrenheit)
-4 -3 -2 -1 0 1 2 3 4

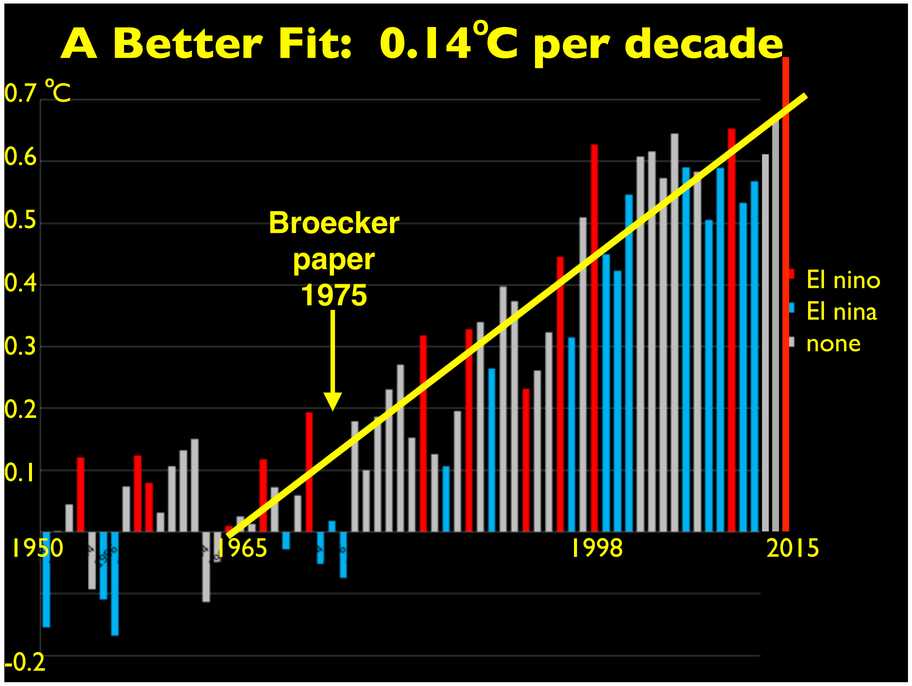

1998
Temperature Difference (Fahrenheit)
-4 -3 -2 -1 0 1 2 3 4

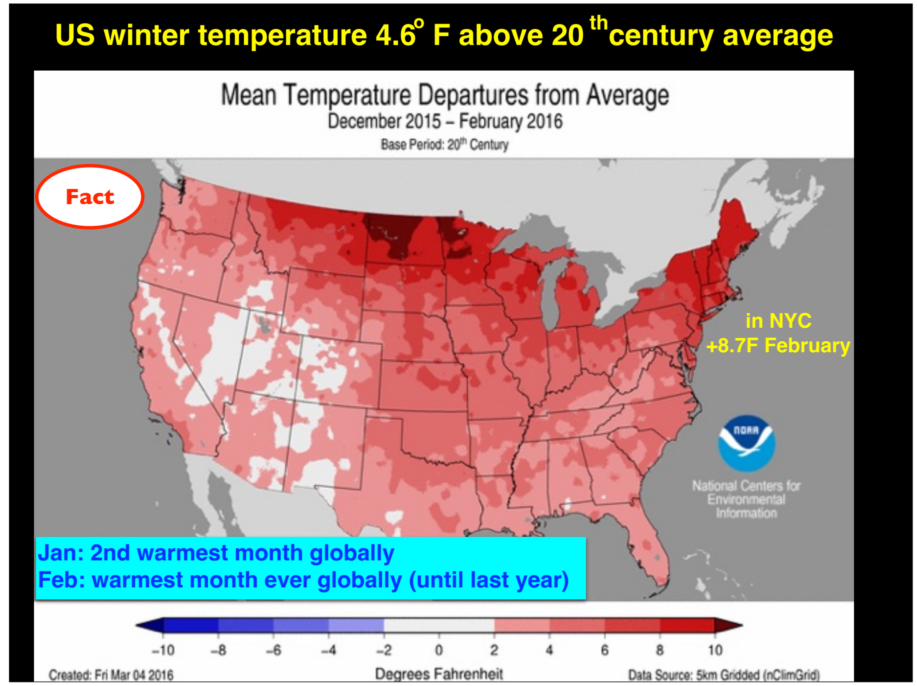

2008
Temperature Difference (Fahrenheit)
-4 -3 -2 -1 0 1 2 3 4

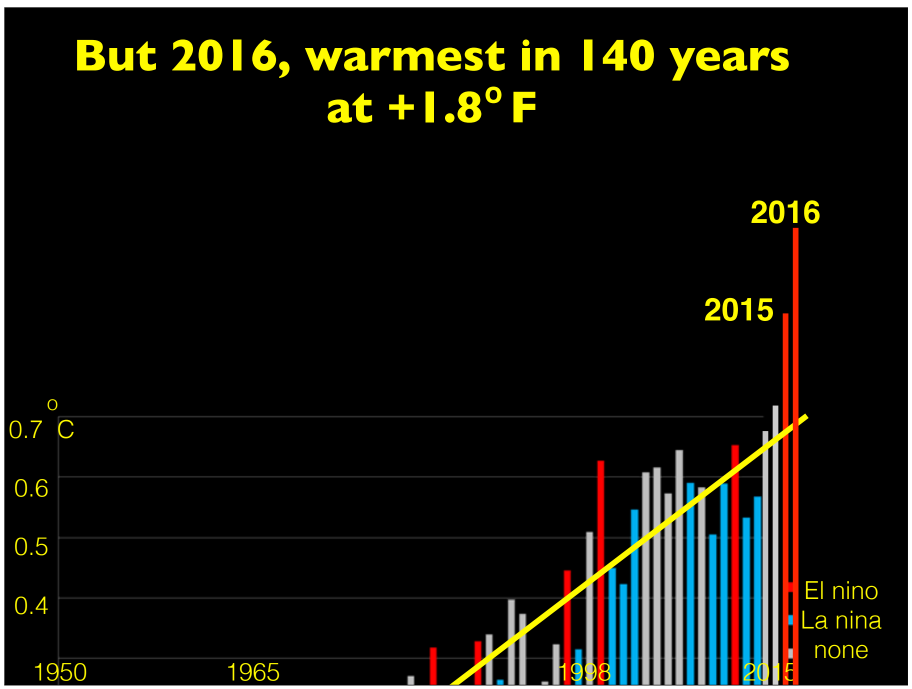

2022
Mean = 58.7F
Temperature Difference (Fahrenheit)
-4 -3 -2 -1 0 1 2 3 4

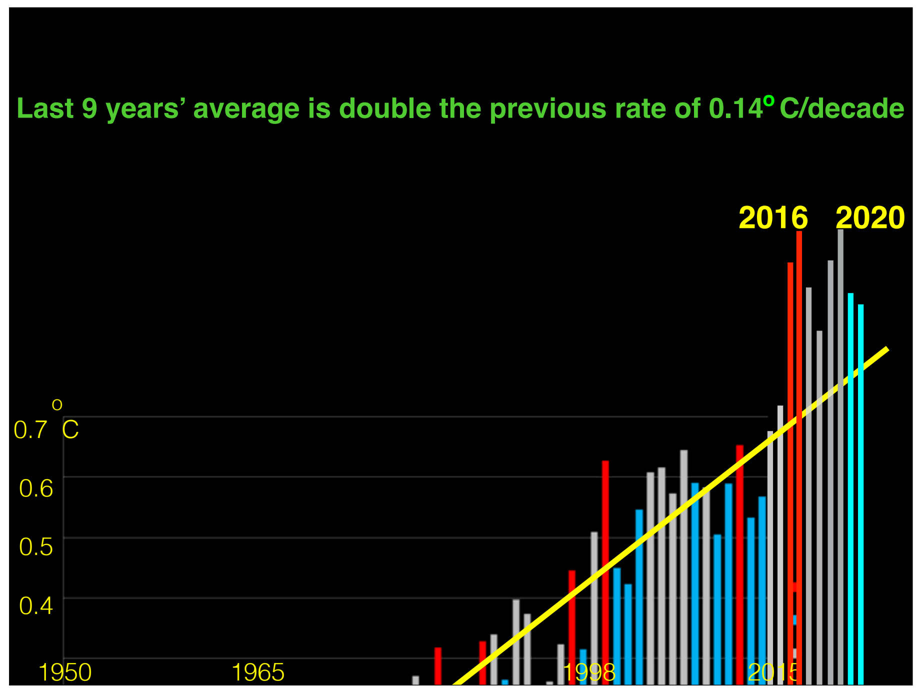

Fact
Average Global Temperature
0.7 °C
0.6
0.5
0.4
0.3
0.2
0.1
-0.2
1950
1976
2015
El nino
La nina
none

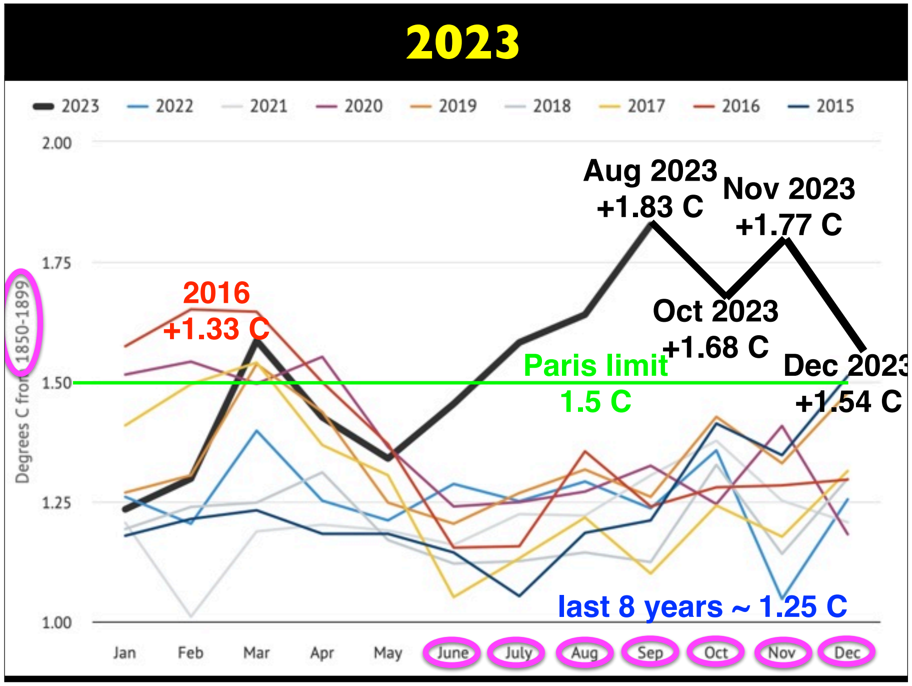

A "plateau" in rising temperatures?
conveniently choose 1997-2012

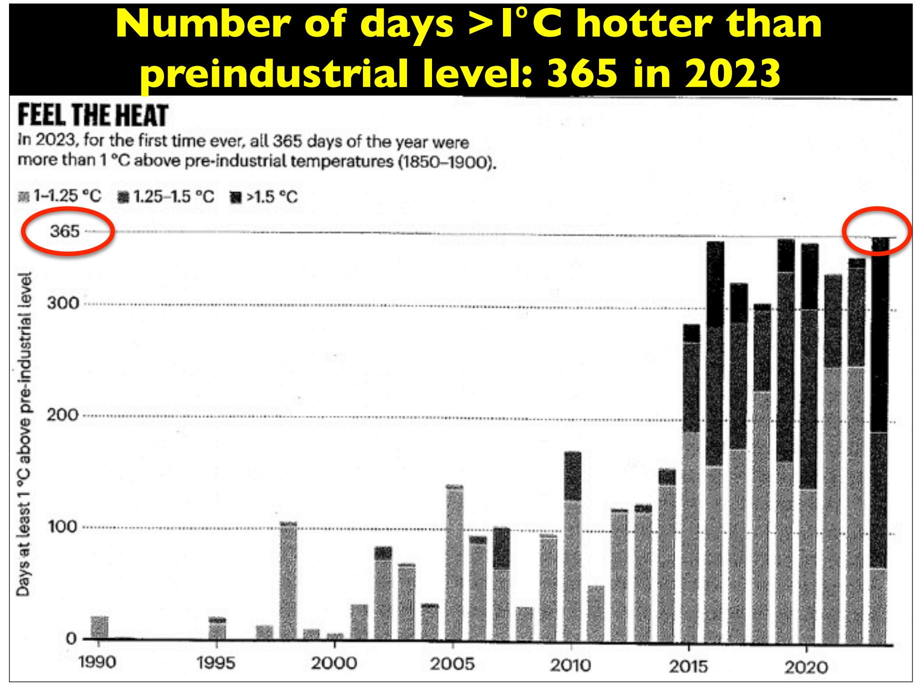

A Better Fit: 0.14°C per decade
0.7 °C
0.6
0.5
0.4
0.3
0.2
0.1
-0.2
Broecker
paper
1975
El nino
El nina
none
1950
1965
1998
2015

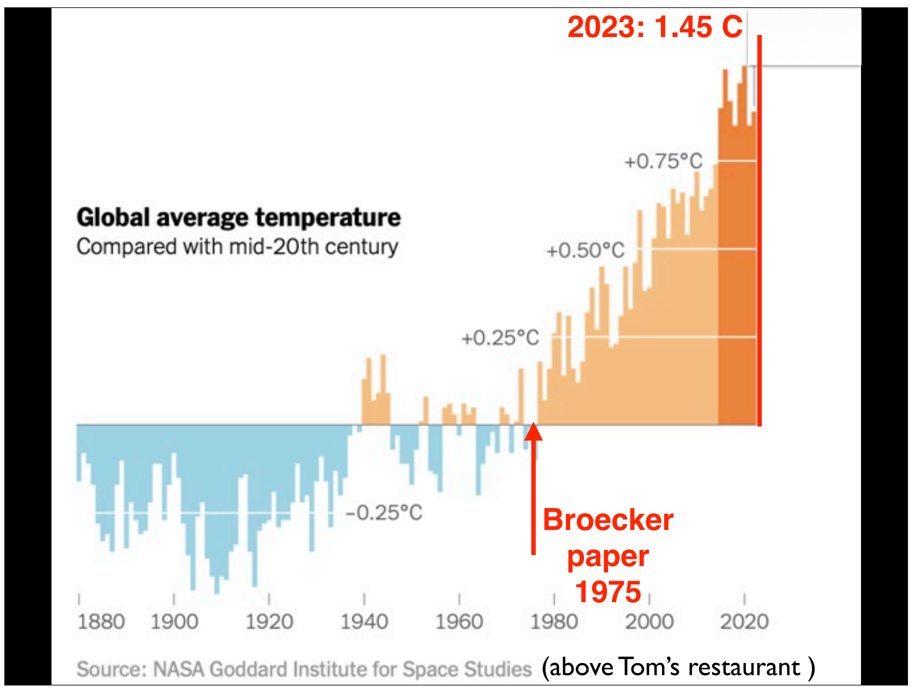

US winter temperature 4.6° F above 20th century average
Mean Temperature Departures from Average
December 2015 – February 2016
Base Period: 20th Century
Fact
in NYC
+8.7F February
Jan: 2nd warmest month globally
Feb: warmest month ever globally (until last year)
National Centers for Environmental Information
−10 −8 −6 −4 −2 0 2 4 6 8 10
Degrees Fahrenheit
Created: Fri Mar 04 2016
Data Source: 5km Gridded (nClimGrid)

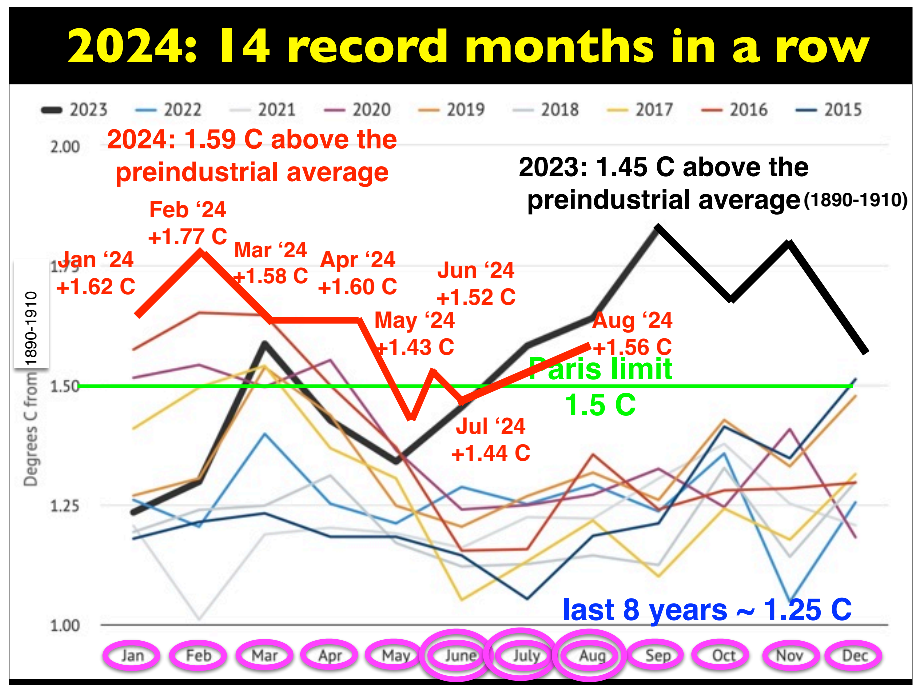

But 2016, warmest in 140 years
at +1.8° F
2016
2015
0.7 °C
0.6
0.5
0.4
El nino
La nina
none
1950
1965
1998
2015


Last 9 years' average is double the previous rate of 0.14° C/decade
2016
2020
0.7 °C
0.6
0.5
0.4
1950
1965
1998
2015


2023
2023 2022 2021 2020 2019 2018 2017 2016 2015
2.00
1.75
1.50
1.25
1.00
Degrees C from 1850-1899
Aug 2023
+1.83 C
Nov 2023
+1.77 C
2016
+1.33 C
Oct 2023
+1.68 C
Paris limit
1.5 C
Dec 2023
+1.54 C
last 8 years ~ 1.25 C
Jan Feb Mar Apr May June July Aug Sep Oct Nov Dec

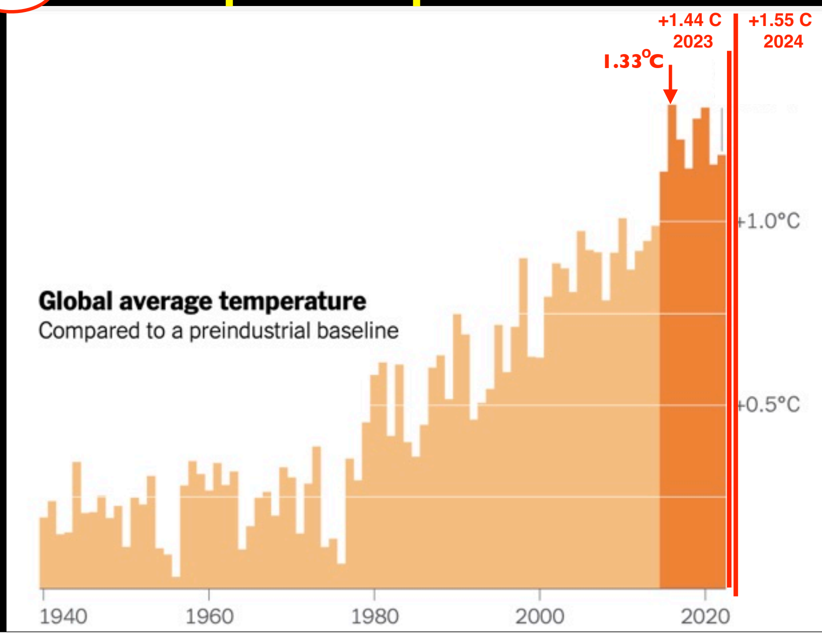

Number of days >1°C hotter than preindustrial level: 365 in 2023
FEEL THE HEAT
In 2023, for the first time ever, all 365 days of the year were more than 1 °C above pre-industrial temperatures (1850–1900).
1–1.25 °C
1.25–1.5 °C
>1.5 °C
365
300
200
100
0
Days at least 1 °C above pre-industrial level
1990 1995 2000 2005 2010 2015 2020


2023: 1.45 C
Global average temperature
Compared with mid-20th century
+0.75°C
+0.50°C
+0.25°C
-0.25°C
Broecker
paper
1975
1880 1900 1920 1940 1960 1980 2000 2020
Source: NASA Goddard Institute for Space Studies (above Tom's restaurant)


2024: 14 record months in a row
2023 2022 2021 2020 2019 2018 2017 2016 2015
2.00
1.75
1.50
1.25
1.00
Degrees C from 1890-1910
2024: 1.59 C above the
preindustrial average
2023: 1.45 C above the
preindustrial average (1890-1910)
Feb '24
+1.77 C
Jan '24
+1.62 C
Mar '24
+1.58 C
Apr '24
+1.60 C
Jun '24
+1.52 C
May '24
+1.43 C
Aug '24
+1.56 C
Paris limit
1.5 C
Jul '24
+1.44 C
last 8 years ~ 1.25 C
Jan Feb Mar Apr May June July Aug Sep Oct Nov Dec

Fact

T compared to pre-industrial T



The temperature record

—global temperatures have been above the 1951-1980 average value for all 47 of the last 47 years (odds this is random fluctuation: 1 in 140 trillion)

—until 2015 the increase was 0.14° C (0.25°F) per decade

—since 2015, all nine years are ~0.2° C (0.36° F) above this extrapolation

—the last 11 yrs are the hottest 11 yrs of the last 150 yrs

—2020 was in a tie as the hottest year in the last 150 yrs

—2024 smashed the record to be the hottest in 150 years

What determines a planet's temperature?

Physics

Energy IN = Energy OUT

Physics

Energy IN = Energy OUT